UNIT - I

1 Introduction to Database Management Systems and ER Model

Syllabus

Introduction, Purpose of Database Systems, Database-System Applications, View of Data, Database Languages, Database System Structure, Data Models. ***Database Design and ER Model :*** *Entity, Attributes, Relationships, Constraints, Keys, Design Process, Entity-Relationship Model, ER Diagram, Design Issues, Extended E-R Features, converting ER and EER diagram into tables.*

Contents

Part I : Introduction to DBMS

1.1 Introduction

- **Definition :** A Database Management System (DBMS) is collection of **interrelated data** and various **programs** that are used to handle the data.
- The **primary goal** of DBMS is to provide a way to **store and retrieve** the required information from the database in convenient and efficient manner.
- For managing the data in the database two important **tasks** are conducted -
  - o **Define the structure** for storage of information.
  - o Provide mechanism for **manipulation of information**.
- In addition, the database systems must ensure the **safety of information** stored.

1.1.1 Characteristics of Database Approach

Following are the characteristics of database system :

1) Representation of some aspects of **real world applications**.
2) Systematic **management of information**.
3) Representing the data by **multiple views**.
4) Efficient and easy implementation of various **operations** such as insertion, deletion and updation.
5) It maintains **data for** some specific **purpose**.
6) It represents **logical relationship** between records and data.

1.2 Purpose of Database Systems

- Earlier database systems are created in response to manage the commercial data. These data is typically stored in files. To allow users to manipulate these files various programs are written for
  1) Addition of new data
  2) Updating the data
  3) Deleting the data.
- As per the addition of new need, separate application programs were required to write. Thus as the time goes by, the system acquires more files and more application programs.
- This typical **file processing system** is supported by conventional operating system. Thus the file processing system can be described as –

- The system that stores the permanent records in files and it needs different application programs to extract or add the records.

1.3 Advantages of DBMS over File Processing Systems

SPPU : May-19, Marks 5

1.3.1 Advantages of DBMS

Following are the advantages of DBMS :

1) DBMS **removes the data redundancy** that means there is no duplication of data in database.
2) DBMS allows to **retrieve the desired data** in required format.
3) **Data** can be **isolated** in separate tables for convenient and efficient use.
4) Data can be **accessed efficiently** using a simple query language.
5) The **data integrity** can be maintained. That means – the constraints can be applied on data and it should be in some specific range.
6) The **atomicity** of data can be maintained. That means, if some operation is performed on one particular table of the database, then the change must be reflected for the entire database.
7) The DBMS allows **concurrent access** to multiple users by using the synchronization technique.
8) The **security policies** can be applied to DBMS to allow the user to access only desired part of the database system.

1.3.2 Disadvantages of DBMS

1) **Complex design :** Database design is complex, **difficult and time consuming**.
2) **Hardware and software cost : Large amount of investment** is needed to setup the required hardware or to repair software failure.
3) **Damaged part :** If one part of database is corrupted or damaged, then **entire database** may get affected.
4) **Conversion cost :** If the current system is in conventional file system and if we need to convert it to database systems then large amount of cost is incurred in purchasing different tools, and adopting different techniques as per the requirement.
5) **Training :** For designing and maintaining the database systems, the people need to be trained.

1.3.3 File Processing System Vs. DBMS

- Earlier database systems are created in response to manage the commercial data. These data is typically stored in files. To allow users to manipulate these files, various programs are written for
  1) Addition of new data
  2) Updating the data
  3) Deleting the data.
- As per the need for addition of new data, separate application programs were required to write. Thus as the time goes by, the system acquires more files and more application programs.
- This typical **file processing system** is supported by conventional operating system. Thus the file processing system can be described as -
- "The system that stores the permanent records in files and it needs different application programs to extract or add the records".
- Before introducing database management system, this file processing system was in use. However, such a system has many drawbacks. Let us discuss them.

Disadvantages of Traditional File Processing System

The traditional file system has following disadvantages :

1) **Data redundancy :** Data redundancy means duplication of data at several places. Since different programmers create different files and these files might have different structures, there are chances that some information may appear repeatedly in some or more format at several places.
2) **Data inconsistency :** Data inconsistency occurs when various copies of same data may no longer get matched. For example changed address of an employee may be reflected in one department and may not be available (or old address present) for other department.
3) **Difficulty in accessing data :** The conventional file system does not allow to retrieve the desired data in efficient and convenient manner.
4) **Data isolation :** As the data is scattered over several files and files may be in different formats, it becomes difficult to retrieve the desired data from the file for writing the new application.
5) **Integrity problems :** Data integrity means data values entered in the database fall within a specified range and are of specific format. With the use of several files enforcing such constraint on the data becomes difficult.

6) **Atomicity problems :** An atomicity means particular operation must be carried out entirely or not at all with the database. It is difficult to ensure atomicity in conventional file processing system.

7) **Concurrent access anomalies :** For efficient execution, multiple users update data simultaneously, in such a case data need to be synchronized. As in traditional file systems, data is distributed over multiple files, one cannot access these files concurrently.

8) **Security problems :** Every user is not allowed to access all the data of database system. Since application program in file system are added in an ad hoc manner, enforcing such security constraints become difficult.

Database systems offer solutions to all the above mentioned problems.

Difference between Database System and Conventional File System

Sr. No.	Database systems	Conventional file systems
1.	Data redundancy is **less**.	Data redundancy is **more**.
2.	**Security** is **high**.	Security is very **low**.
3.	Database systems are used when security **constraints are high**.	Conventional file systems are used where there is **less demand** for **security constraints**.
4.	Database systems define the data in a **structured** manner. Also there is well defined co-relation among the data.	File systems define the data in **un-structured** manner. Data is usually in isolated form.
5.	Data **inconsistency** is **less** in database systems.	Data **inconsistency** is **more** in file systems.
6.	**User** is **unknown** to the physical address of the data used in database systems.	User locates the **physical address of file** to access the data in conventional file systems.
7.	We can retrieve the data in any **desired format** using database systems.	We cannot retrieve the data in any desired format using file systems.
8.	There is ability to access the data **concurrently** using database systems.	There is **no ability to concurrently** access the data using conventional file system.

Review Question

1. *Define DBMS. Explain advantages of DBMS over file system.*

SPPU : May-19, End Sem, Marks 5

1.4 Database-System Applications

There are wide range of applications that make use of database systems. Some of the applications are -

1) **Accounting :** Database systems are used in maintaining information employees, salaries, and payroll taxes.
2) **Manufacturing :** For management of supply chain and tracking production of items in factories database systems are maintained.
3) For maintaining customer, product and purchase information the databases are used.
4) **Banking :** In banking sector, for customer information, accounts and loan and for performing banking applications the DBMS is used.
5) For purchase on credit cards and generation of monthly statements database systems are useful.
6) **Universities :** The database systems are used in universities for maintaining student information, course registration, and accounting.
7) **Reservation systems :** In airline / railway reservation systems, the database is used to maintain the reservation and schedule information.
8) **Telecommunication :** In telecommunications for keeping records of the calls made, generating monthly bills, maintaining balances on prepaid calling cards, and storing information about communication networks the database systems are used.

1.5 View of Data

SPPU : May-19, Oct.-19, Marks 5

- Database is a collection of interrelated data and set of programs that allow users to access or modify the data.
- Abstract view of the system is a view in which the system hides certain details of how the data are stored and maintained.
- The **main purpose** of database systems is to provide users with abstract view of the data.
- The view of the system help the user to **retrieve data** efficiently.
- For simplifying the user interaction with the system there are several levels of abstraction - these levels are - Physical level, logical level and view level.

Review Question

1. *Elaborate the need of database views. Also explain the situations where in the view created are updateable views.* **SPPU : May-19, End Sem, Oct.-19, In Sem, Marks 5**

1.6 Data Abstraction

SPPU : Dec.-17, May-19, Marks 5

Definition of data abstraction : Data abstraction means retrieving only the required amount of information about the system and **hiding** background details.

There are several levels of abstraction that simplify user interactions with the system. These are :

1) Physical level :

- This is the **lowest level.**
- This level describes how the data are **stored.**
- The database administrators decide how to store data at the physical level.
- This level describes complex low-level data structures.

2) Logical level :

- This is the next **higher level,** which describes what data are stored in the database?.
- This level also describes the **relationship between the data.**
- The logical level thus describes the entire database in terms of a small number of relatively simple structures.
- The database administrators use a logical level of abstraction for deciding what information to keep in the database.

3) View level :

- This is the **highest level** of abstraction that describes only part of the entire database.
- The view level can provide access to only part of the database.
- This level helps in **simplifying the interaction** with the system.
- It can provide **multiple views** of the same system.
- For example - A Clerk at the reservation system can see only part of the database and access the passenger's required information.

Fig. 1.6.1 shows the relationship between the three levels of abstraction.

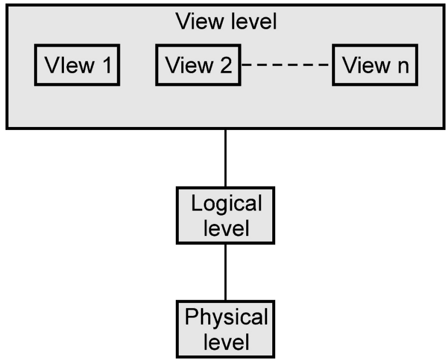


Fig. 1.6.1 : Levels of data abstraction

For example : Consider following record

```
type employee = record
        empID : numeric(10)
        empname : char(20)
        dept_no : numeric(10)
        salary : numeric(8,2)
        end
```

This code defines a new record **employee** with four fields. Each field is associated with field name and its **type**. There are several other records such as **department** with fields dept_no, dept_name, building customer with fields cust_id,cust_name

- **At the physical level**, the record - **customer, employee, department** can be described as block of **consecutive storage locations**. Many database systems hide lowest level storage details from database programmer.
- The type definition of the records is decided **at the logical level**. The **programmer** work of the record at this level, similarly **database administrators** also work at this level of abstraction.
- There is specific view of the record is allowed at the view level. For instance - customer can view the name of the employee, or id of the employee but cannot access employee's salary.

Review Question

1. *For the database system to be usable, it must retrieve data efficiently. The need of efficiency has led designers to use complex data structures to represent data in the database. Developers hides this complexity from the database system users through several levels of abstraction. Explain those levels of abstraction in detail.* **SPPU : Dec.-17, May-19, End Sem, Marks 5**

1.7 Database Languages

There are three types of languages supported by database systems.

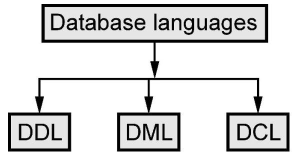


Fig. 1.7.1 Types of database languages

(1) DDL

- Data Definition Language (DDL) is a specialized language used to specify a database schema by a set of definitions.
- It is a language used for **creating** and **modifying** the structures of tables, views, indexes, etc.
- DDL is also used to specify additional properties of data.
- Some of the common commands used in DDL are -**CREATE, ALTER, DROP.**
- The **primary use** of CREATE command is to build a new table. Using ALTER command, the users can add up some additional column and drop existing columns. Using DROP command, the user can delete table or view.

(2) DML

- DML stands for Data Manipulation Language.
- This language enables users to access or manipulate data as organized by appropriate data model.
- The types of access are -
  - o **Retrieval** of information stored in the database
  - o **Insertion** of new information into the database.
  - o **Deletion** of information from the database.
  - o **Modification** of information stored in database.
- There are two types of DML -
  - o **Procedural DML** - Require a user to specify what data are needed and how to get those data.
  - o **Declarative DML** - Require a user to specify what data are needed without specifying how to get those data.
- **Query** is a statement used for requesting the retrieval of information. This retrieval of information using some specific language is called **query language.**

(3) DCL

- The Data Control Language (DCL) is used to control access to data stored in the database. This is also called as authorization.
- The typical command used in DCL are GRANT and REVOKE.
  - **GRANT :** This command is used to give access rights or privileges to the database.
  - **REVOKE :** The revoke command removes user access rights or privileges to the database objects

1.8 Database System Structure

SPPU : May-18, Oct.-18, Dec.-18,19, Marks 5

1.8.1 Overall Structure of DBMS

Three Schema Architecture

- **Definition :** Database schema is a collection of database objects like tables, views, indexes and so on associated with one particular database username. This username is called the **schema owner**.
- For example Student Schema can be owner of STUDENT and MARKS tables. The Course schema can be the owner of SUBJECT table.
- The **goal** of three-schema architecture is to separate the user application from the physical database.
- The architecture of database is divided into **three levels** based on three types of schema - internal schema, conceptual schema or external schema.

1. Internal level :

- It contains **internal schema**.
- This schema represents the **physical storage structure of database**.
- This schema is maintained by the software and user is not allowed to modify it.
- This level is closest to the physical storage. It typically describes the record layout of the files and types of files, access paths etc.

2. Conceptual level :

- It contains **conceptual schema**.
- This schema **hides the details** of internal level.
- This level is also called as logical level as it contains the constructs used for designing the database.

- It contains information like table name, their columns, indexes and constraints, database operations.
- A representational data model is used to describe conceptual schema when a database system is implemented.

3. External level :

- It contains the external schema or user views.
- At this level, the user will get to see only the data stored in the database. Either they will see whole data values or any specific records. They will not have any information about how they are stored in the database.

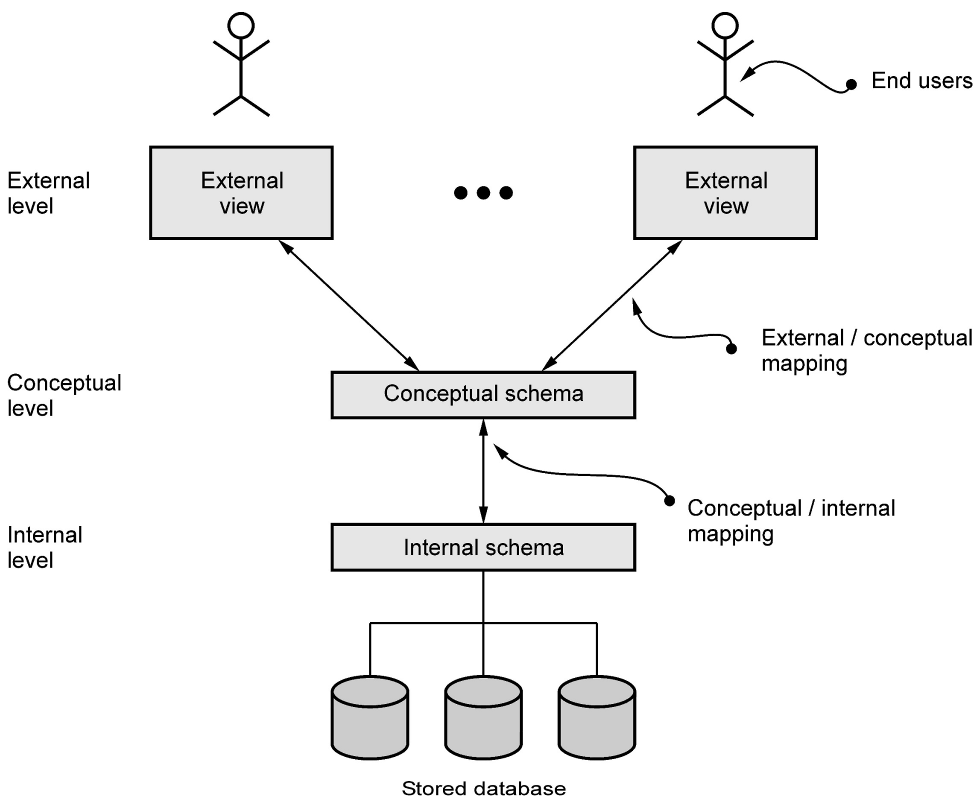


Fig. 1.8.1 Three schema architecture

- The processes of transforming requests and results between levels are called **mappings**.
- In the three schema architecture there are two mappings –

  1) External - Conceptual Mapping and

  2) Conceptual - Internal Mapping

1.8.2 Architecture of DBMS

- The typical structure of typical DBMS is based on relational data model as shown in Fig. 1.8.2.

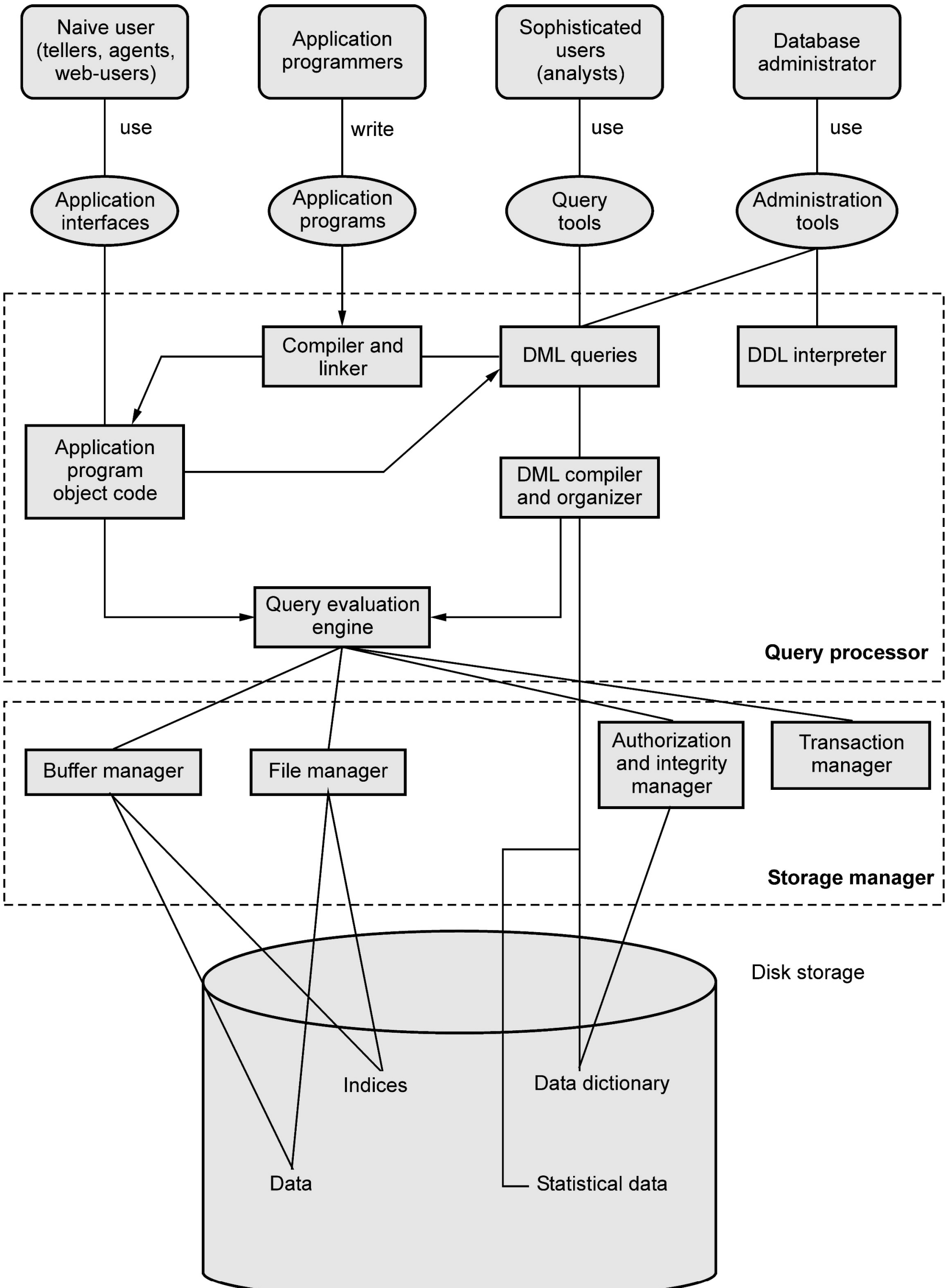


Fig. 1.8.2 Architecture of database

- Consider the top part of Fig. 1.8.2. It shows **application interfaces** used by naïve users, application programs created by application programmers, query tools used by sophisticated users and administration tools used by database administrator
- The lowest part of the architecture is for **disk storage.**
- The two important components of database architecture are - **Query processor and storage manager.**

Query processor :

- The interactive query processor helps the database system to simplify and facilitate access to data. It consists of DDL interpreter, DML compiler and query evaluation engine.
- With the following components of query processor, various functionalities are performed -
  - i) **DDL interpreter :** This is basically a translator which interprets the DDL statements in data dictionaries.
  - ii) **DML compiler :** It translates DML statements query language into an evaluation plan. This plan consists of the instructions which query evaluation engine understands.
  - iii) **Query evaluation engine :** It executes the low-level instructions generated by the DML compiler.
- When a user issues a query, the parsed query is presented to a query optimizer, which uses information about how the data is stored to produce an efficient execution plan for evaluating the query. An execution plan is a blueprint for evaluating a query. It is evaluated by **query evaluation engine**.

Storage Manager :

- o Storage manager is the component of database system that provides interface between the low level data stored in the database and the application programs and queries submitted to the system.
- o The storage manager is responsible for storing, retrieving, and updating data in the database. The storage manager components include -
  - i) **Authorization and integrity manager :** Validates the users who want to access the data and tests for integrity constraints.
  - ii) **Transaction manager :** Ensures that the database remains in consistent despite of system failures and concurrent transaction execution proceeds without conflicting.
  - iii) **File manager :** Manages allocation of space on disk storage and representation of the information on disk.

iv) Buffer manager : Manages the fetching of data from disk storage into main memory. The buffer manager also decides what data to cache in main memory. Buffer manager is a crucial part of database system.

- Storage manager implements several data structures such as -

  i) Data files : Used for storing database itself.

  ii) Data dictionary : Used for storing metadata, particularly schema of database.

  iii) Indices : Indices are used to provide fast access to data items present in the database

Review Questions

1. *Draw and explain overall structure of database system.*

   SPPU : May-18, Dec.-18, End Sem, Marks 5

2. *Different components of database management systems like query processor, storage manager, transaction manager etc. are functional for processing the query submitted by user. Explain the functions of each component in view of getting query output.*

   SPPU : Oct.-18, In Sem, Marks 5

3. *Draw the overall database system structure. Explain storage manager, transaction manager and query processor in detail.*

   SPPU : Dec.-19, End Sem, Marks 5

1.9 Data Models

SPPU : Nov.-19, Marks 5

- **Definition :** It is a collection of conceptual tools for describing data, relationships among data, semantics (meaning) of data and constraints.
- Data model is **a structure below the database**.
- Data model provides a way to describe the design of database at physical, logical and view level.
- There are various data models used in database systems and these are as follows -

(1) Relational model :

- The relation model consists of **collection of tables** which stores data and also represents the relationship among the data.
- The **table** is also known as **relation.**
- The table contains one or more **columns** and each column has unique name.
- Each table contains **record of particular type,** and each record type defines a fixed **number of fields or attributes.**

- **For example –** The following figure shows the relational model by showing the relationship between Student and Result database. For example – Student **Ram** lives in city **Chennai** and his marks are **78.** Thus the relationship between these two databases is maintained by the **SeatNo.** Column

Seat No	Name	City
101	Ram	Chennai
102	Shyam	Pune

SeatNo	Marks
101	78
102	95

Advantages :

(i) Structural independence : Structural independence is an ability that allows us to make changes in one database structure without affecting other. The relational model have structural independence. Hence making required changes in the database is convenient in relational database model.

(ii) Conceptual simplicity : The relational model allows the designer to simply focus on logical design and not on physical design. Hence relational models are conceptually simple to understand.

(iii) Query capability : Using simple query language (such as SQL) user can get information from the database or designer can manipulate the database structure.

(iv) Easy design, maintenance and usage : The relational models can be designed logically hence they are easy to maintain and use.

Disadvantages :

i) Relational model **requires powerful hardware** and large data storage devices.

ii) May lead to **slower processing time.**

iii) Poorly designed systems lead to **poor implementation** of database systems.

(2) Entity relationship model :

- As the name suggests the entity relationship model uses collection of basic objects called **entities** and **relationships**.
- The entity is a thing or object in the real world.
- The entity relationship model is widely used in database design.
- For example - Following is a representation of Entity Relationship model in which the relationship **works_for** is between entities **Employee** and **Department.**

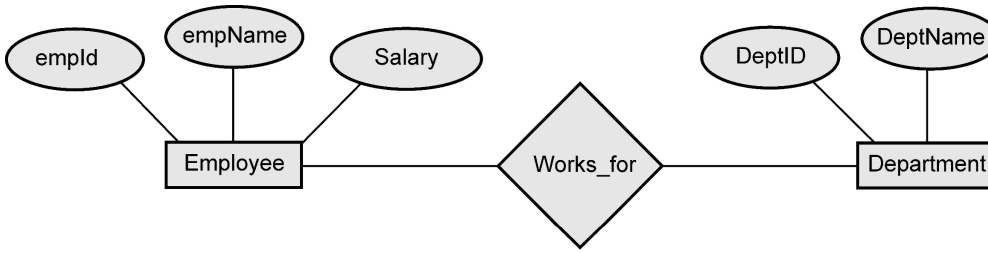


Fig. 1.9.1

Advantages :

i) **Simple :** It is simple to draw ER diagram when we know entities and relationships.

ii) **Easy to understand :** The design of ER diagram is very logical and hence they are easy to design and understand.

iii) **Effective :** It is effective communication tool.

iv) **Integrated :** The ER model can be easily integrated with Relational model.

v) **Easy conversion :** ER model can be converted easily into other type of models.

Disadvantages :

i) **Loss of information :** While drawing ER model some information can be hidden or lost.

ii) **Limited relationships :** The ER model can represent limited relationships as compared to other models.

iii) **No representation for data manipulation :** It is not possible to represent data manipulation in ER model.

iv) **No industry standard :** There is no industry standard for notations of ER diagram.

(3) Object Based Data Model :

- The **object oriented languages** like C++, Java, C# are becoming the dominant in software development.
- This led to object based data model.
- The object based data model combines **object oriented features** with relational data model.

Advantages :

i) **Enriched modelling :** The object based data model has capability of modelling the real world objects.

ii) **Reusability :** There are certain features of object oriented design such as inheritance, polymorphism which help in reusability.

iii) **Support for schema evolution :** There is a tight coupling between data and applications, hence there is strong support for schema evolution.

iv) **Improved performance :** Using object based data model there can be significant improvement in performance using object based data model.

Disadvantages :

i) **Lack of universal data model :** There is no universally agreed data model for an object based data model, and most models lack a theoretical foundation.

ii) **Lack of experience :** In comparison with relational database management the use of object based data model is limited. This model is more dependent on the skilled programmer.

iii) **Complex :** More functionalities present in object based data model make the design complex.

(4) Semi-structured data model :

- The semi-structured data model permits the **specification of data** where individual data items of same type may have **different sets of attributes.**
- The Extensible Markup Language **(XML)** is widely used to represent semi-structured data model.

Advantages

i) Data is not constrained by fixed schema.

ii) It is **flexible.**

iii) It is **portable.**

Disadvantage

i) **Queries** are **less efficient** than other types of data model.

(5) Hierarchical Model

- In this model each entity has only **one parent** but can have **several children.** At the top of hierarchy there is only one node called **root**. Refer Fig. 1.9.2.
- This model represents the relationship in 1 : N types. That means one university can have multiple courses. One course can have multiple projects and so on.

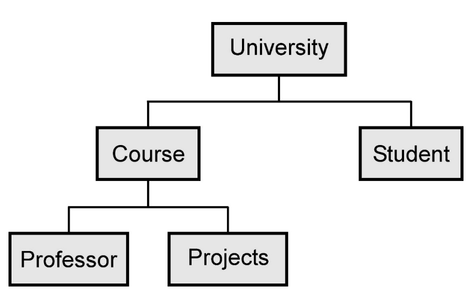


Fig. 1.9.2 Hierarchical model

Advantage

1. This model groups the data into tables and defines the relationship between the tables.

Disadvantages

1. For searching any data, we have to start form the root and move downwards and visit each child node. Thus traversing through each node is required.
2. For addition of some information about child node, sometimes the parent information need to be modified.
3. It fails to handle many to many relationship (M : N) efficiently.

   It can cause duplication and data redundancy.

(6) Network Model

- This is enhanced version of hierarchical model. It overcomes the drawback of hierarchical model. It helps to address M : N relationship. That means, this model is not having single parent concept. Any child in this model can have multiple parents. Refer Fig. 1.9.3.
- The **main difference** between network model and hierarchical model is to allow many to many relationship.

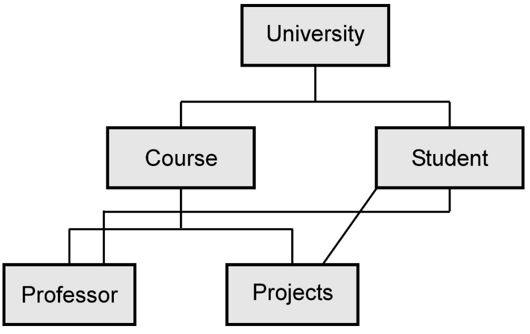


Fig. 1.9.3 Network model

Advantages

1. **Capability to handle more Relationships :** Since the network model allows many to many relationship, it helps in modeling the real life situations.
2. **Ease of data access :** The data access is easier and flexible than hierarchical model.
3. **Data Integrity :** In network model every member is associated with some other member in the model.
4. **Conformance to Standards :** The network model structure can be designed as per the standards.

Disadvantages

1. **Complex to implement :** For all the records the pointers need to be maintained, hence the database structure becomes complex.
2. **Complicated Operations :** The simple operations such as insertion, deletion and modification becomes complex due to adjustment of multiple pointer.

3. **Difficult to change structure :** The structural changes are difficult.

1.10 Data Independence

SPPU : Nov.-19, Oct.-19, Aug.-17,Marks 5

- **Definition :** Data independence is an ability by which one can change the data at one level without affecting the data at another level. Here level can be **physical, conceptual** or **external.**
- Data independence is one of the important characteristics of database management system.
- By this property, the structure of the database or the values stored in the database can be easily modified by without changing the application programs.
- There are two types of data independence :

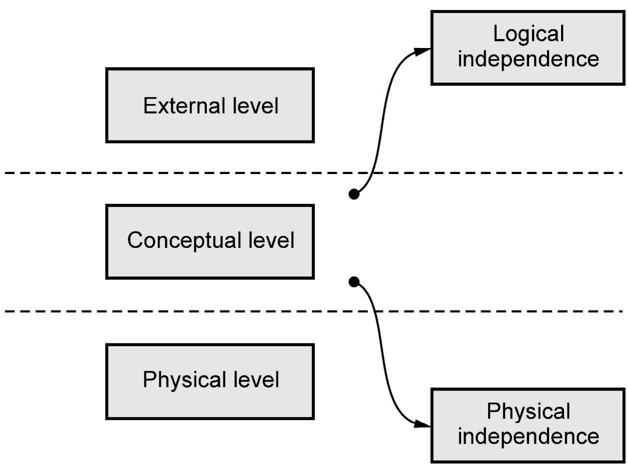


Fig. 1.10.1 Data independence

1. **Physical Independence :** This is a kind of data independence which allows the modification of physical schema without requiring any change to the conceptual schema. For example - if there is any change in memory size of database server then it will not affect the logical structure of any data object.
2. **Logical Independence :** This is a kind of data independence which allows the modification of conceptual schema without requiring any change to the external schema. For example - Any change in the table structure such as addition or deletion of some column does not affect user views.

- By these data independence the time and cost acquired by changes in any one level can be reduced and abstract view of data can be provided to the user.

Review Question

1. *Explain with example what is physical data independence. Also explain its importance.*

SPPU : Aug.-17, In Sem, Marks 5

1.11 Database Users

SPPU : May 19, Nov.-18, Marks 5

There **four different types** of database system **users** differentiated by the way they interact with the system. Different types of user interfaces for different types of users are -

i) **Naïve users :** This type of users interact with the system with the help of previously created program(known as application program). Typically a form interface is used by this type of user to interact with the system. For example - we may feel up the booking form for booking a ticket on an online system.

ii) **Application programmers :** These are computer professionals who write application programs. Normally Rapid Application Development (RAD) tools are used to quickly design forms and reports.

iii) **Sophisticated users :** These are the type of users who interact with the system without writing programs. These users may submit the database query to retrieve the desired information or tools from data analysis software. Database analysts fall in this category of database users.

iv) **Specialized users :** Specialized users are sophisticated users who write specialized database application that does not fit into the traditional data-processing framework. Among these applications are computer aided-design systems, knowledge-base and expert systems etc.

Responsibilities of DBA

Database Administrator (DBA) is a person who have a **central control** over both data and programs that access data in DBMS. The **functions of DBA** are -

i) **Schema definition :** DBA creates a database Schema using DDL statements.

ii) **Schema and physical organization modification :** In order to improve the overall performance of database management system DBA carried out changes in schema or physical organization.

iii) **Granting authorization for data access :** Granting authorization for data access means giving special permissions to the users for accessing the database. This task is done by DBA so that privacy of database can be maintained.

iv) **Maintenance :** Maintenance involve different types of tasks such as monitoring jobs and their performances, taking periodic back-ups, making free space available for normal operations or upgrading disk space as per the requirements. These all tasks are carried out by DBA.

Example 1.11.1 *Explain the problems that may arrive if the DBA does not discharge the responsibilities properly.* **SPPU : May 19, End Sem, Marks 5**

Solution : Following are the problems that may arrive if DBA does not discharge the responsibilities properly -

1) The database can not perform without file manger interaction. If nothing is stored in the files then obviously we can not retrieve anything.
2) The consistency in database must operations must be maintained. If it is not, then it will create major problems. For instance – account balance may go below the minimum allowed, employees can earn too much overtime and so on.
3) If authorization for the authentic user is not done, then unauthorized users may access the database or users authorized to access part of the database may be able to access parts of the database for which they lack authority.
4) Data can be lost permanently.
5) Consistency constraints may be violated despite proper integrity enforcement in each transaction. For example, incorrect bank balances might be reflected due to simultaneous withdrawals and deposits, and so on.

Part II: Data Modelling

1.12 Database Design and ER Model

SPPU : May-18, Nov.-17,19, Marks 5

- Data Modelling in database management system is based on Entity Relationship modelling(ER Model)
- Entity Relational model is a model for identifying entities to be represented in the database and representation of how those entities are related.
- ER data model represents the overall **logical structure** of database.
- The E-R model is very useful in mapping the meanings and interactions of real-world entities onto a conceptual schema or database.
- The ER model consists of three basic concepts –

1. Entities 2. Relationships 3. Attributes

1.12.1 Entity and Entity Sets

- **Entity :** An entity is an object that exists and is distinguishable from other objects. For example - Student named "Poonam" is an entity and can be identified by her name. The entity can be concrete or abstract. The concrete entity can be - Person, Book, Bank. The abstract entity can be like - holiday, concept entity is represented as a box.

- **Entity set :** The entity set is a set of entities of the same types. For example - All students studying in class X of the School. The entity set need not be disjoint. Each entity in entity set have the same set of attributes and the set of attributes will distinguish it from other entity sets. No other entity set will have exactly the same set of attributes.

1.12.2 Relationships

- **Relationship:** Relationship is an association among two or more entities.
- **Relationship Set:** The **relationship set** is a collection of similar relationships. For example - Following Fig. 1.12.1 shows the relationship **works_for** for the two entities **Employee** and **Departments.**

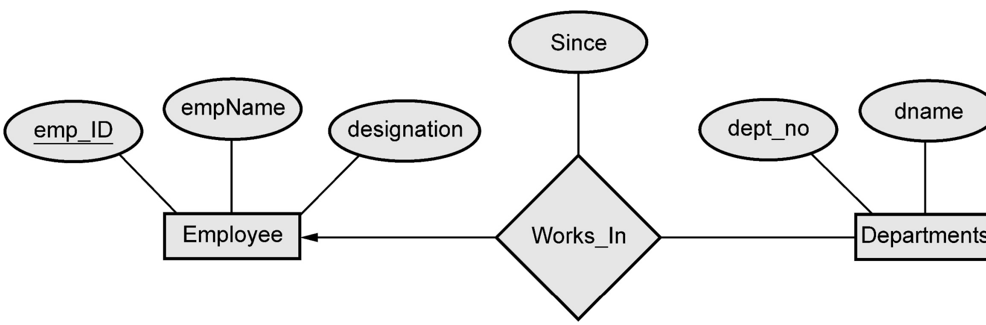


Fig. 1.12.1 : Relation set

- The association between entity sets is called as **participation**. that is, the entity sets E1, E2, . . . , En participate in relationship set R.
- The function that an entity plays in a relationship is called that entity's **role**.

1.12.3 Attributes

- Attributes define the properties of a data object of entity. For example: if student is an entity, his ID, name, address, date of birth, class are its attributes. The attributes help in determining the unique entity. Refer Fig. 1.12.2 for Student entity set with attributes - ID, name, address. Note that entity is shown by rectangular box and attributes are shown in oval. The primary key is underlined.

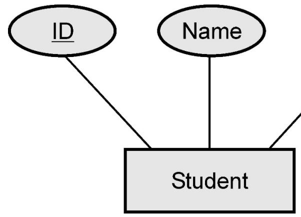


Fig. 1.12.2 : Student entity set with attributes

1.12.3.1 Types of Attributes

Following are the types of attributes -

1) Simple and Composite Attributes :

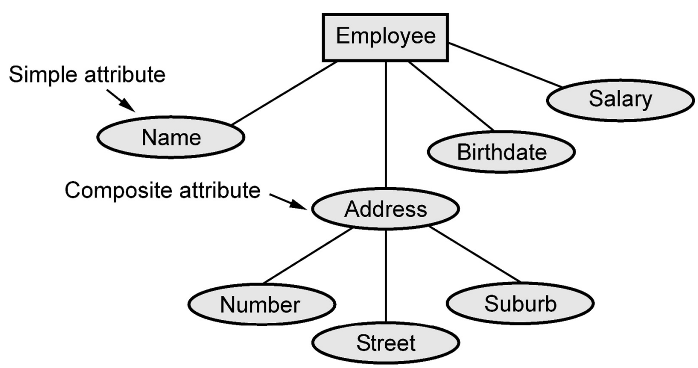
1) Simple attributes are attributes that are drawn from the atomic value domains

   For example - Name = {Parth} ; Age = {23}

2) Composite attributes: Attributes that consist of a hierarchy of attributes

   For example – Address may consists of "Number", "Street" and "Suburb"

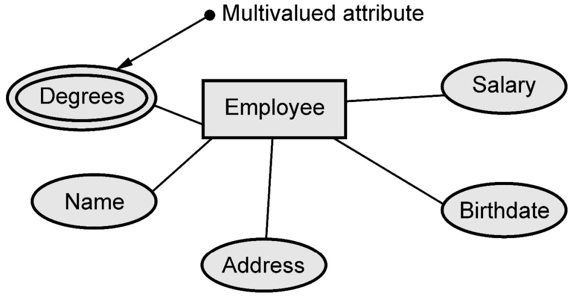
   Hence, **Address = {59 + 'JM Road' + 'ShivajiNagar'}**



2) Single valued and multivalued :

- There are some attributes that can be represented using a single value. For example - StudentID attribute for a Student is specific only one studentID.
- Multivalued attributes : Attributes that have a set of values for each entity. It is represented by concentric ovals

  For example - Degrees of a person: ' BSc' , 'MTech', 'PhD'



3) Derived attribute :

Derived attributes are the attributes that contain values that are calculated from other attributes. To represent derived attribute there is dotted ellipse inside the solid ellipse. For example – Age can be derived from attribute DateOfBirth. In this situation, DateOfBirth might be called Stored Attribute.

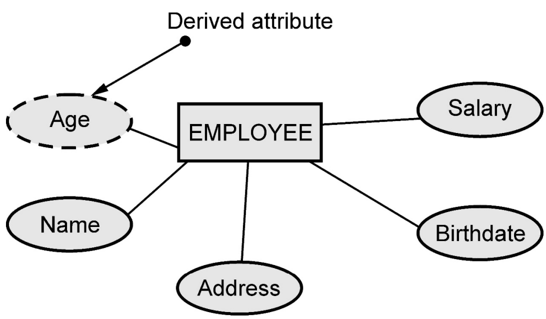


Fig. 1.12.3

1.13 Constraints

- Relationship types have certain rules that limit the possible combination of entities that can take part in relationship. These rules or restrictions are called **structural constraints.**
- The common type of structural constraint is represented by the **cardinality ratio**.
- The **cardinality ratio** for a binary relationship specifies the **maximum** number of relationship instances that an entity can participate in.

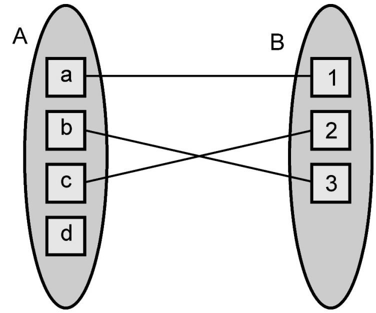
1.13.1 Types of Cardinality

Mapping Cardinality represents the number of entities to which another entity can be associated via a relationship set.

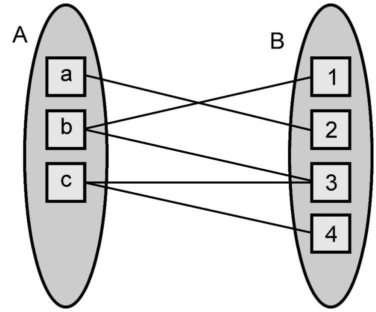
The mapping cardinalities are used in representing the binary relationship sets.

Various types of mapping cardinalities are -

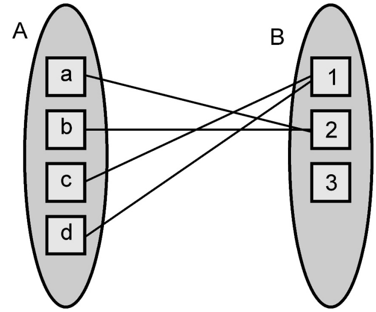
1. **One to One :** An entity A is associated with at least one entity on B and an entity B is associated with at one entity on A. This can be represented as



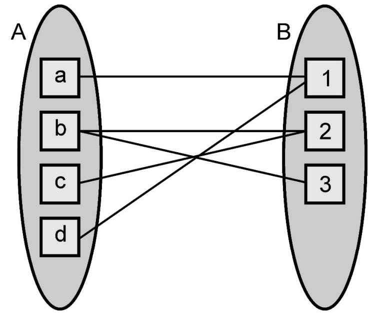
2. **One to Many :** An entity in A is associated with any number of entities in B. An entity in B, however, can be associated with at most one entity in A.



3. **Many to One :** An entity in A is associated with at most one entity in B. An entity in B, however, can be associated with any number of entities in A.



4. **Many to many :** An entity in A is associated with any number (zero or more) of entities in B, and an entity in B is associated with any number (zero or more) of entities in A.



1.14 Keys

SPPU : May-19, Oct.-19, Marks 5

- Keys are used to identify entities uniquely from the given entity set.
- A key can be a an attribute or a set of attributes that help us to identify the entity uniquely.
- Keys also help to identify relationships uniquely, and thus distinguish relationships from each other.
- The primary key of an entity set allows us to distinguish among the various entities of the set.
- For example - If a Student table contains the information about various students as given below -

RollNo	Name	City	Course
101	Ram	Pune	Computer
102	Sita	Pune	Electronics
103	Laxman	Chennai	Mechanical

In above table, **RollNo** is a primary key because, it uniquely identifies the student record.

Review Question

1. *Distinguish between super key, candidate key and primary key.*

SPPU : May-19, End Sem, Oct.-19, In Sem, Marks 5

1.15 Design Process

Following are the six steps of database design process. The ER model is most relevant to first three steps

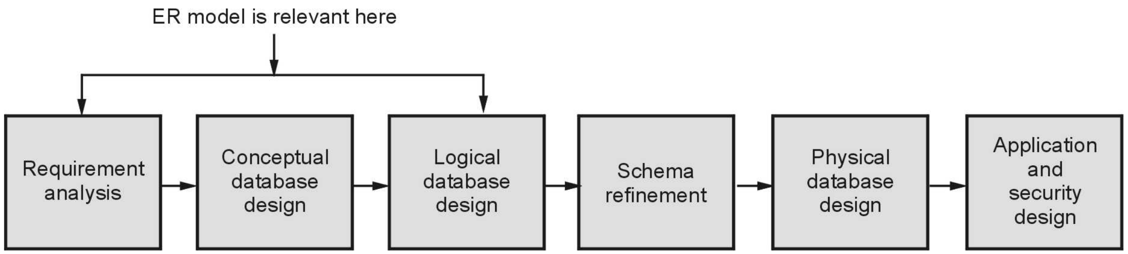


Fig. 1.15.1 : Database design process

Step 1 : Requirement analysis :

- In this step, it is necessary to understand what data need to be stored in the database, what applications must be built, what are all those operations that are frequently used by the system.
- The requirement analysis is an **informal** process and it requires proper **communication** with user groups.
- There are several methods for **organizing and presenting information** gathered in this step.
- Some automated tools can also be used for this purpose.

Step 2 : Conceptual database design :

- This is a steps in which **E-R Model** i.e. Entity Relationship model is built.
- E-R model is a **high level data model** used in database design.
- The **goal** of this design is to create a simple description of data that matches with the requirements of users.

Step 3 : Logical database design :

- This is a step in which ER model in **converted to relational database schema,** sometimes called as the logical schema in the relational data model.

Step 4 : Schema refinement :

- In this step, **relational database schema is analyzed** to identify the potential problems and to refine it.
- The schema refinement can be done with the help of **normalizing and restructuring the relations.**

Step 5 : Physical database design :

- In this step, the design of database is **refined further.**

- The tasks that are performed in this step are - building **indexes** on tables and **clustering** tables, redesigning some parts of schema obtained from earlier design steps.

Step 6 : Application and security design :

- Using design methodologies like UML(Unified Modeling Language) the design of the database can be accomplished.
- The **role of each entity** in every process must be reflected in the application task.
- For each role, there must be the provision for **accessing** the some part of database and **prohibition of access** to some other part of database.
- Thus some **access rules** must be enforced on the application(which is accessing the database) to protect the **security features.**

1.16 ER Diagram

SPPU : Aug.-17, Marks 2

- Entity Relational model is a model for identifying entities tobe represented in the database and representation of how those entities are related.
- The ER data model specifies enterprise schema that represents the overall **logical structure** of a database graphically.
- E-R diagrams are used to model real-world objects like a person, a car, a company and the relation between these real-world objects.

Features of ER model

i) E-R diagrams are used to represent E-R model in a database, which makes them easy to be converted into relations (tables).

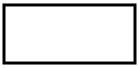
ii) E-R diagrams provide the purpose of real-world modeling of objects which makes them intently useful.

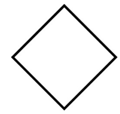
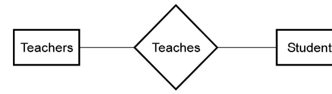
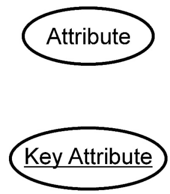
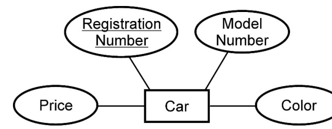
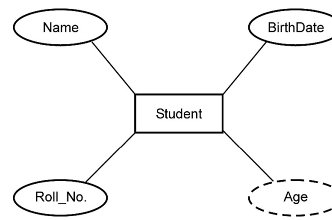
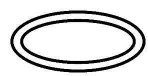
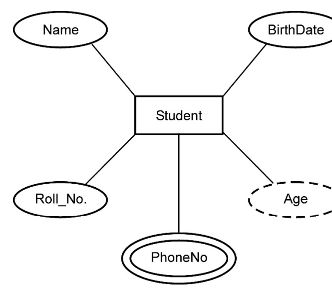
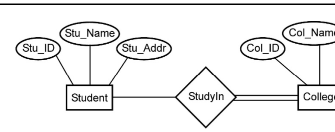
iii) E-R diagrams require **no technical knowledge** and no hardware support.

iv) These diagrams are very **easy to understand** and easy to create even by a naive user.

v) It gives a standard solution of **visualizing** the **data logically.**

Various Components used in ER Model are -

Component	Symbol	Example
Entity : Any real-world object can be represented as an entity about which data can be stored in a database. All the real world objects like a book, an organization, a		

product, a car, a person are the examples of an entity.		
Relationship : Rhombus is used to setup relationships between two or more entities.		
Attribute : Each entity has a set of properties. These properties of each entity are termed as attributes. For example, a car entity would be described by attributes such as price, registration number, model number, color etc		
Derived attribute : Derived attributes are those which are derived based on other attributes, for example, age can be derived from date of birth. To represent a derived attribute, another dotted ellipse is created.		
Multivalued attribute : An attribute that can hold multiple values is known as multivalued attribute. We represent it with double ellipses in an E-R Diagram. E.g. A person can have more than one phone numbers so the phone number attribute is multivalued.		
Total participation : Each entity is involved in the relationship. Total participation is represented by double lines.		

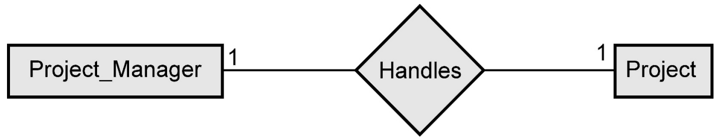
1.17 Conventions

SPPU : Oct.-19, Marks 2

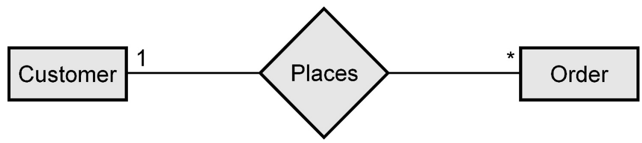
1.17.1 Mapping Cardinality Representation

There are four types of relationships that are considered for key constraints.

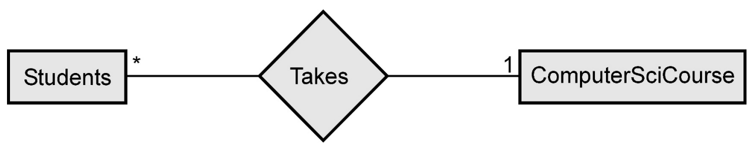
i) **One to one relation :** When entity A is associated with at the most one entity B then it shares one to one relation. For example - There is one project manager who manages only one project.



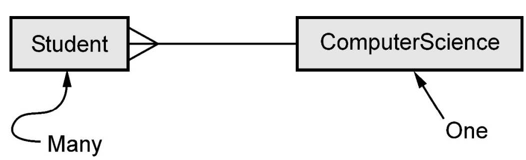
ii) One to many : When entity A is associated with more than one entities at a time then there is one to many relation. For example - One customer places order at a time.



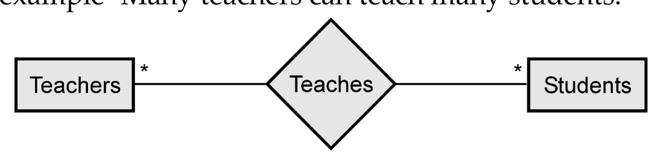
iii) Many to one : When more than one entities are associated with only one entity then there is is many to one relation. For example - Many student take a ComputerSciCourse.



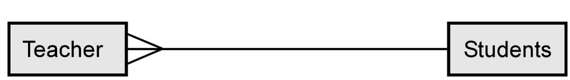
Alternate representation can be



iv) Many to many : When more than one entities are associated with more than one entities. For example -Many teachers can teach many students.

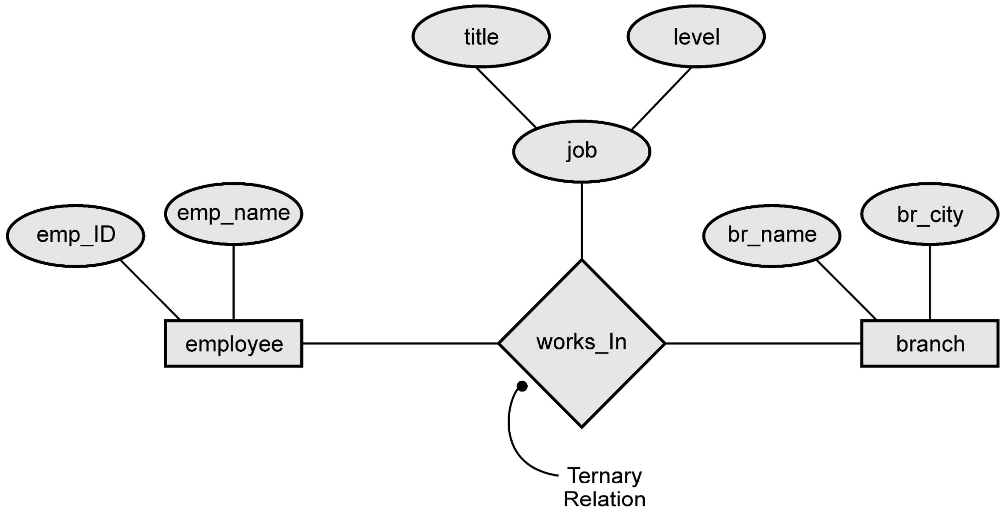


Alternate representation can be



1.17.2 Ternary Relationship

The relationship in which three entities are involved is called ternary relationship. For example -



1.17.3 Weak Entity Set

- A weak entity is an entity that cannot be uniquely identified by its attributes alone. The entity set which does not have sufficient attributes to form a primary key is called as weak entity set.

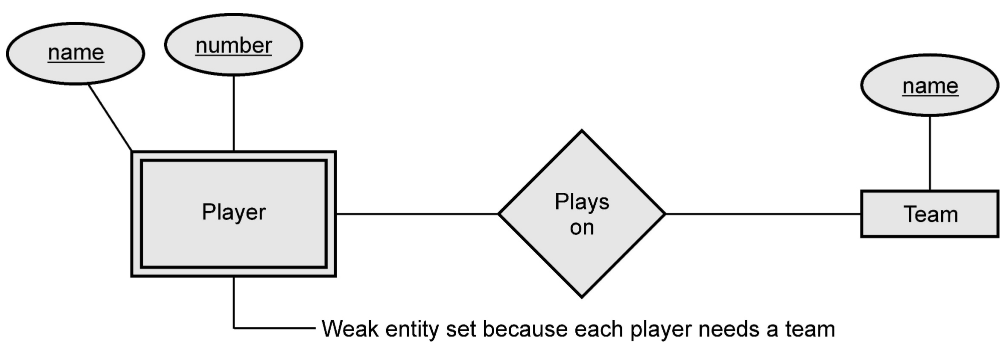


Fig. 1.17.1 : Weak entity set

- **Strong Entity Set**

  The entity set that has primary key is called as strong entity set

Weak entity rules

- A weak entity set has **one or more many-one** relationships to other (supporting) entity sets.
- The key for a weak entity set is its own underlined attributes and the keys for the supporting entity sets. For example - player-number and team-name is a key for Players.

Difference between Strong and Weak Entity Set

Sr. No.	Strong entity set	Weak entity set
1	It has its own primary key.	It does not have sufficient attribute to form a primary key on its own.
2.	It is represented by rectangle	It is represented by double rectangle.
3.	It represents the primary key which is underlined.	It represents the partial key or discriminator which is represented by dashed underline.
4.	The member of strong entity set is called as dominant entity set	The member of weak entity set is called subordinate entity set.
5.	The relationship between two strong entity sets is represented by diamond symbol.	The relationship between strong entity set and weak entity set is represented by double diamond symbol.
6.	The primary key is one of the attributes which uniquely identifies its member.	The primary key of weak entity set is a combination of partial key and primary key of the strong entity set.

1.18 Design Issues

- The ER diagram can be designed in several different ways for representing the same system application. Different representations may not always be exactly equivalent.

(1) Use of Entity Vs. Attribute

- Use of entity or attribute in the ER diagram depends upon the real-world application. For example - Following is an example, in which phone_number is represented as attribute. But we can represent phone_number as entity instead of attribute. The advantage of such representation is that it is possible to maintain the multiple phone numbers for a customer. Moreover, we can maintain some extra information such as customer's **location** if the phone_number is represented as entity instead of attribute.

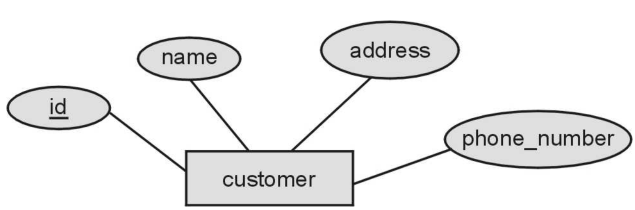


Fig. 1.18.1 : Representation of Phone_number as attribute

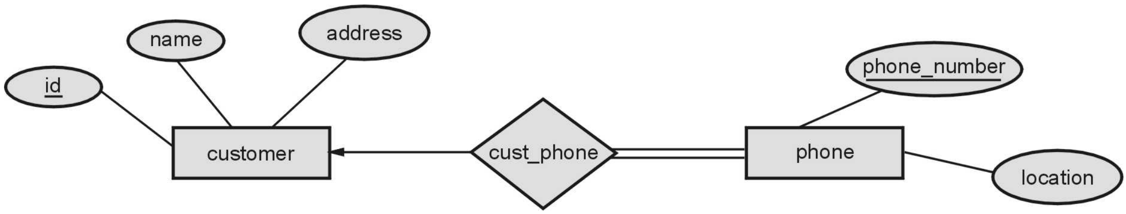
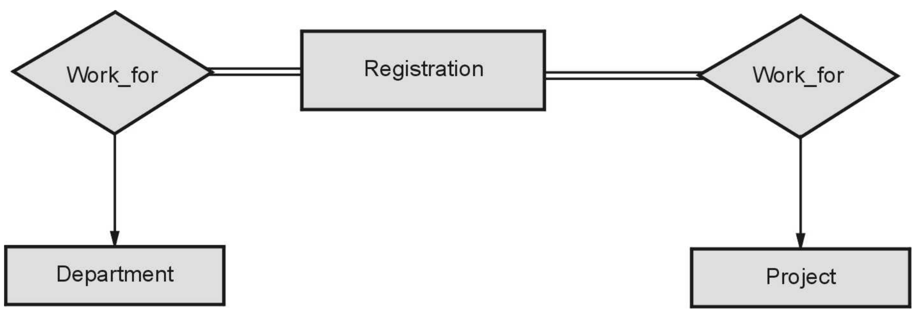


Fig. 1.18.2 : Representation of Phone_number as entity

(2) Use of Entity Set Vs. Relationship Sets

- Representing a particular as entity set or relationship is a common problem, many times designer face. For example as shown in the following figure , an – Employee works for a Department and an Employee works for a project, These two relationships can be represented by making Registration as an entity set and representing works_for relationship set for both department and project.

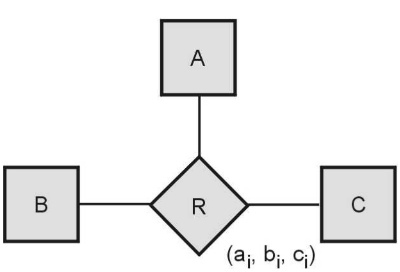


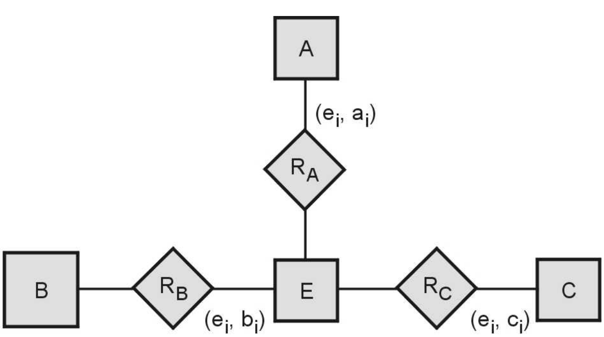
(3) Binary Vs. n-ary Relationship Sets

Generally, the relationships described in the databases are binary relationships. However, non-binary relationships can be represented by several binary relationships.

We can specify some mapping cardinalities on relationships with degree > 2

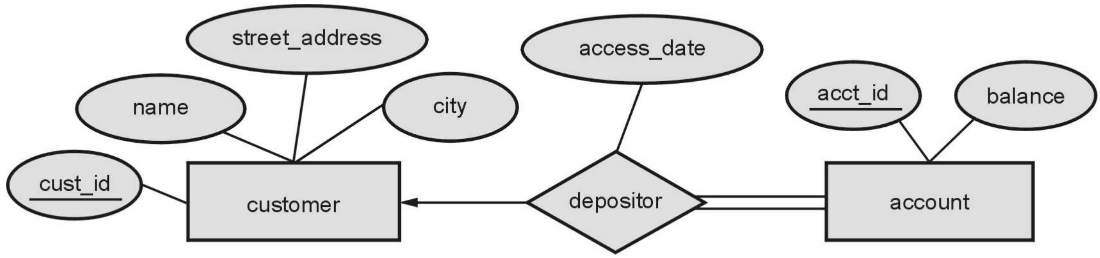
For example –





(4) Placing Relationship Attributes

- The relationship set can have descriptive attribute. For example – in the following fig. the relationship depositor has descriptive attribute named **access_date.** The decision of placing the specified attribute as a relationship or entity attribute should possess the characteristics of the real world enterprise that is being modelled.



1.19 Extended E-R Features

1.19.1 Specialization and Generalization

- Some entities have relationships that form hierarchies. For instance, Employee can be an hourly employee or contracted employee.
- In this relationship hierarchies, some entities can act as superclass and some other entities can act as subclass.
- **Superclass :** An entity type that represents a general concept at a high level, is called superclass.
- **Subclass :** An entity type that represents a specific concept at lower levels, is called subclass.
- The subclass is said to inherit from superclass. When a subclass inherits from one or more superclasses, it inherits all their attributes. In addition to the inherited attributes, a subclass can also define its own specific attributes.
- The process of making subclasses from a general concept is called **specialization.** This is **top-down** process. In this process, the sub-groups are identified within an entity set which have attributes that are not shared by all entities.
- The process of making superclass from subclasses is called **generalization.** This is a **bottom up** process. In this process multiple sets are synthesized into high level entities.
- The symbol used for specialization/ Generalization is

- **For example –** There can be two subclass entities namely **Hourly_Emps** and **Contract_Emps** which are subclasses of **Empoyee** class. We might have attributes **hours_worked** and **hourly_wage** defined for **Hourly_Emps** and an attribute **contractid** defined for **ContractEmps**.

  Therefore, the attributes defined for an **Hourly_Emps** entity are the attributes for **Employees** plus **Hourly_Emps**. We say that the attributes for the entity set **Employees** are inherited by the entity set **Hourly_Emps** and that **Hourly-Emps ISA** (read is a) Employees. It can be represented by following Fig. 1.19.1.

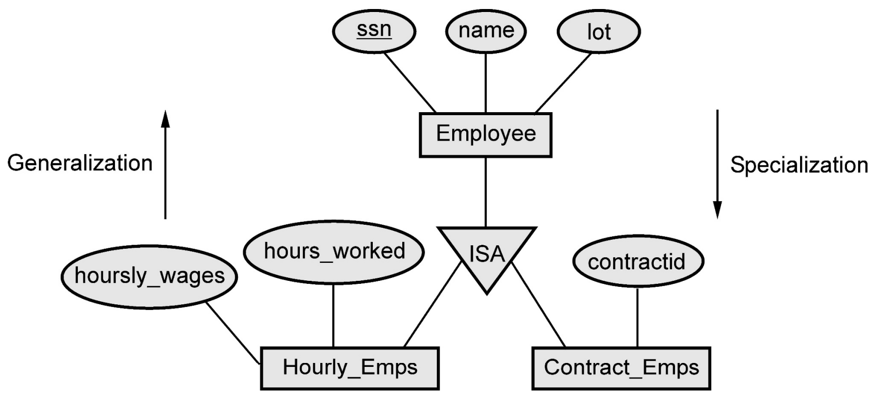


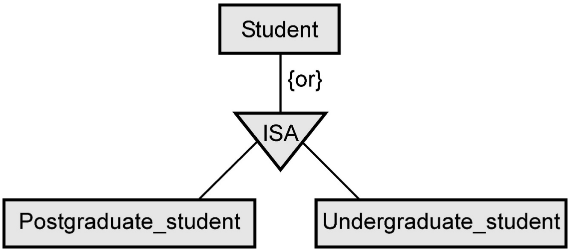
Fig. 1.19.1 Example of Generalization and Specialization

1.19.2 Constraints on Specialization / Generalization

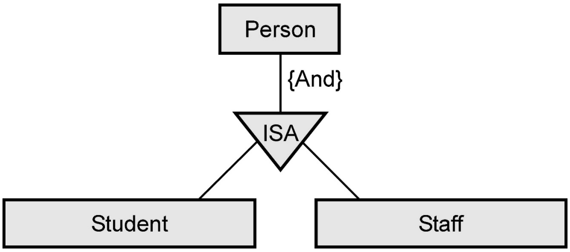
There are four types of constraints on specialization/generalization relationship. These are -

1) **Membership constraints :** This is a kind of constraints that involves determining which entities can be members of a given lower-level entity. There are two types of membership constraints -
   i) **Condition defined :** In condition-defined lower-level entity sets,membership is evaluated on the basis of whether or not an entity satisfies an explicit condition or predicate. For example - Consider the high-level entity Set **Employee** that has attribute **Employee_type**. All Employee entities are evaluated on defining **Employee_type** attribute. All entities that satisfy the condition student type = "ContractEmployee" are included in Contracted Employee. Since all the lower-level entities are evaluated on the basis of the same attribute this type of generalization is said to be **attribute-defined**.
   ii) **User defined :** This is kind of entity set that in which the membership is manually defined.
2) **Disjoint constraints :** The disjoint constraint only applies when a superclass has

more than one subclass. If the subclasses are disjoint, then an entity occurrence can be a member of only one of the subclasses. For entity Student has either **Postgraduate_Student** entity or **Undergraduate_Student**

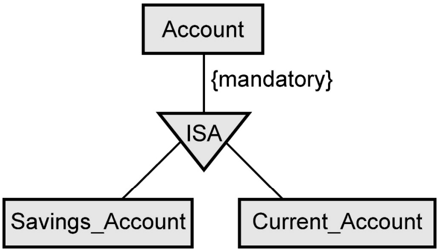


3) **Overlapping :** When some entity can be a member of more than one subclasses. For example - Person can be both a Student or a Staff. The **And** can be used to represent this constraint.

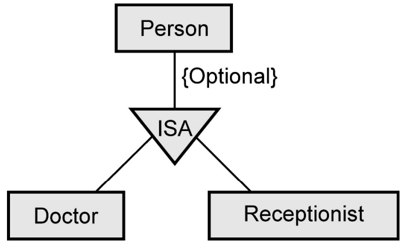


4) **Completeness :** It specifies whether or not an entity in the higher-level entity set must belong to at least one of the lower-level entity sets within the generalization/specialization. This constraint may be one of the following -

   i) **Total generalization or specialization :** Each higher-level entity must belong to a lower-level entity set. For example - Account in the bank must either Savings account or Current Account. The **mandatory** can be used to represent this constraint.



   ii) **Partial generalization or specialization :** Some higher-level entities may not belong to any lower-level entity set.



1.19.3 Aggregation

A feature of the entity relationship model that allows a relationship set to participate in another relationship set. This is indicated on an ER diagram by drawing a dashed box around the aggregation.

For example - We treat the relationship set work and the entity sets employee and project as a higher-level entity set called work.

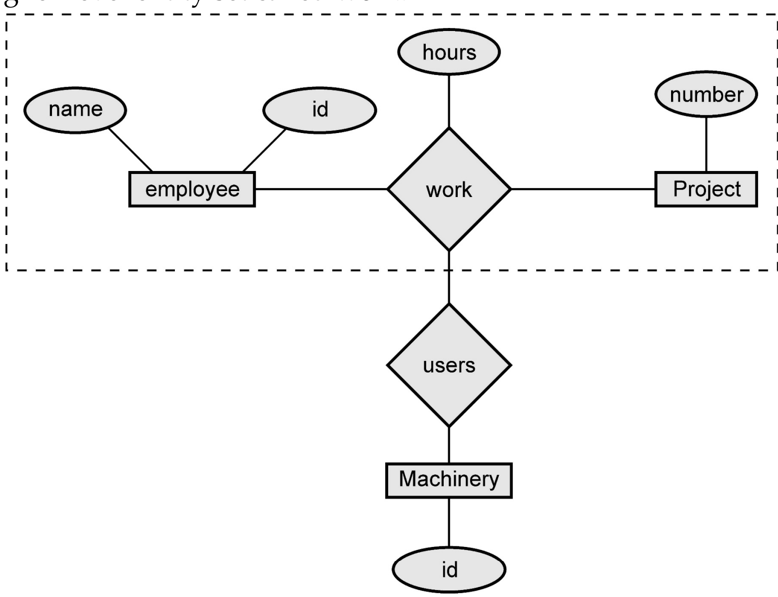


Fig. 1.19.2 : ER model with aggregation

1.20 Converting ER and EER Diagram into Tables

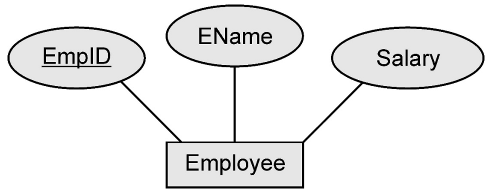
SPPU : Nov.-18, Marks 4

In this section we will discuss how to map various ER model constructs to Relational Model construct.

1.20.1 Mapping of Entity Set to Relationship

- An entity set is mapped to a relation in a straightforward way.
- Each attribute of entity set becomes an attribute of the table.
- The primary key attribute of entity set becomes an entity of the table.

- For example - Consider following ER diagram.



The converted employee table is as follows -

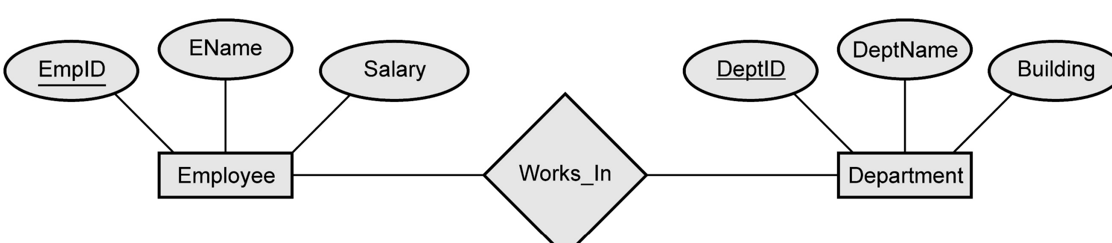
EmpID	EName	Salary
201	Poonam	30000
202	Ashwini	35000
203	Sharda	40000

The SQL statement captures the information for above ER diageam as follows -

```
CREATE TABLE Employee( EmpID CHAR(11),
    EName CHAR(30),
    Salary INTEGER,
PRIMARY KEY(EmpID))
```

1.20.2 Mapping Relationship Sets(without Constraints) to Tables

- Create a table for the relationship set.
- Add all primary keys of the participating entity sets as fields of the table.
- Add a field for each attribute of the relationship.
- Declare a primary key using all key fields from the entity sets.
- Declare foreign key constraints for all these fields from the entity sets.
  For example - Consider following ER model

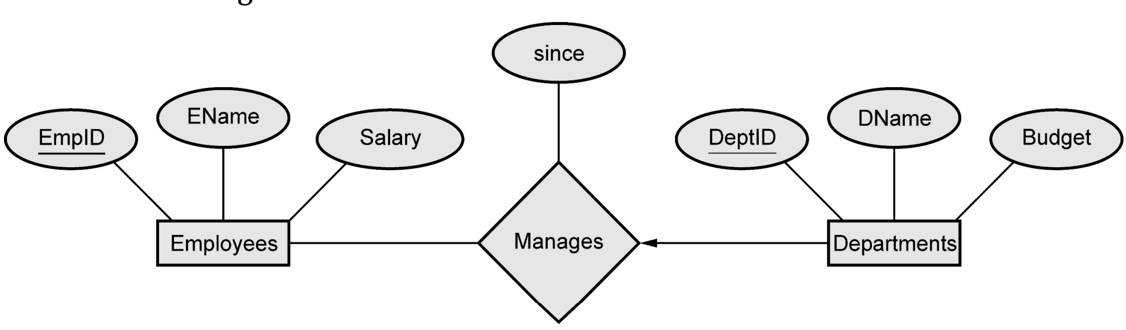


The SQL statement captures the information for relationship present in above ER diagram as follows -

```
CREATE TABLE Works_In  (EmpID CHAR(11),
                        DeptID CHAR(11),
                        EName CHAR(30),
                        Salary INTEGER,
                        DeptName CHAR(20),
                        Building CHAR(10),
                        PRIMARY KEY(EmpID,DeptID),
                        FOREIGN KEY (EmpID) REFERENCES Employee,
                        FOREIGN KEY (DeptID) REFERENCES Department
                        )
```

1.20.3 Mapping Relationship Sets(With Constraints) to Tables

- If a relationship set involves **n** entity sets and some **m** of them are linked **via arrows** in the ER diagram, the key for anyone of these **m** entity sets **constitutes a key** for the relation to which the relationship set is mapped.
- Hence we have **m** candidate keys, and one of these should be designated as the **primary key.**
- There are two approaches used to convert a relationship sets with key constraints into table.
- **Approach 1 :**
  - o By this approach the relationship associated with more than one entities is separately represented using a table. For example - Consider following ER diagram. Each **Dept has at most one manager, according to the key constraint on Manages.**



Here the constraint is each department has at the most one manager to manage it. Hence no two tuples can have same DeptID. Hence there can be a separate table named **Manages** with DeptID as Primary Key. The table can be defined using following SQL statement

```
CREATE TABLE Manages(EmpID CHAR(11),
                    DeptID INTEGER,
                    Since DATE,
                    PRIMARY KEY(DeptID),
                    FOREIGN KEY (EmpID) REFERENCES Employees,
                    FOREIGN KEY (DeptID) REFERENCES Departments)
```

- **Approach 2 :**
  - o In this approach , it is preferred **to translate a relationship set with key constraints**.
  - o It is a superior approach because, it avoids creating a distinct table for the relationship set.
  - o The idea is to include the information about the relationship set in the table corresponding to the entity set with the key, taking advantage of the key constraint.
  - o This approach eliminates the need for a separate Manages relation, and queries asking for a department's manager can be answered without combining information from two relations.
  - o The only drawback to this approach is that space could be wasted if several departments have no managers.
  - o The following SQL statement, defining a **Dep_Mgr** relation that captures the information in both **Departments** and **Manages**, illustrates the second approach to translating relationship sets with key constraints :

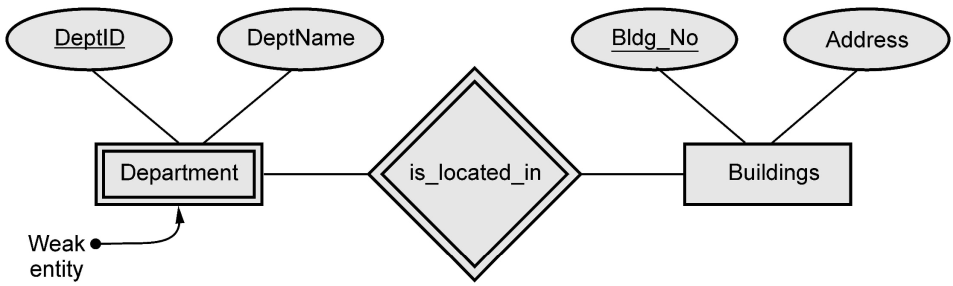
```
CREATE TABLE Dep_Mgr ( DeptID INTEGER,
                    DName CHAR(20),
                    Budget REAL,
                    EmpID CHAR (11),
                    since DATE,
                    PRIMARY KEY (DeptID),
                    FOREIGN KEY (EmpID) REFERENCES Employees)
```

1.20.4 Mapping Weak Entity Sets to Relational Mapping

A weak entity can be identified uniquely only by considering the primary key of another (owner) entity. Following steps are used for mapping Weka Entity Set to Relational Mapping

- Create a table for the weak entity set.
- Make each attribute of the weak entity set a field of the table.
- Add fields for the primary key attributes of the identifying owner.
- Declare a foreign key constraint on these identifying owner fields.
- Instruct the system to automatically delete any tuples in the table for which there are no owners

For example - Consider following ER model

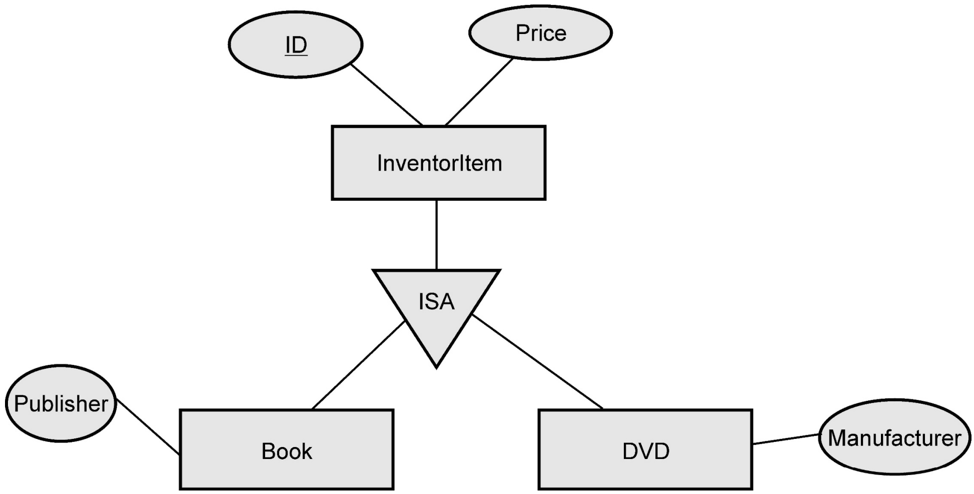


Following SQL Statement illustrates this mapping

```
CREATE TABLE Department(DeptID CHAR(11),
        DeptName CHAR(20),
        Bldg_No CHAR(5),
        PRIMARY KEY (DeptID,Bldg_No),
        FOREIGN KEY(Bldg_No) References Buildings on delete cascade
        )
```

1.20.5 Mapping of Specialization / Generalization(EER Construct) to Relational Mapping

The specialialization/Generalization relationship(Enhanced ER Construct) can be mapped to database tables(relations) using three methods. To demonstrate the methods, we will take the – InventoryItem, Book, DVD



Method 1 : All the entities in the relationship are mapped to individual tables

InventoryItem(ID , Price)
Book(ID,Publisher)
DVD(ID, Manufacturer)

Method 2 : Only subclasses are mapped to tables. The attributes in the superclass are duplicated in all subclasses. For example -

Book(ID, Price,Publisher)
DVD(ID, Price,Manufacturer)

Method 3 : Only the superclass is mapped to a table. The attributes in the subclasses are taken to the superclass. For example -

InventoryItem(ID , Price,Publisher,Manufacturer)

This method will introduce **null** values. When we insert a **Book** record in the table, the **Manufacturer** column value will be null. In the same way, when we insert a **DVD** record in the table, the **Publisher** value will be null.

Review Question

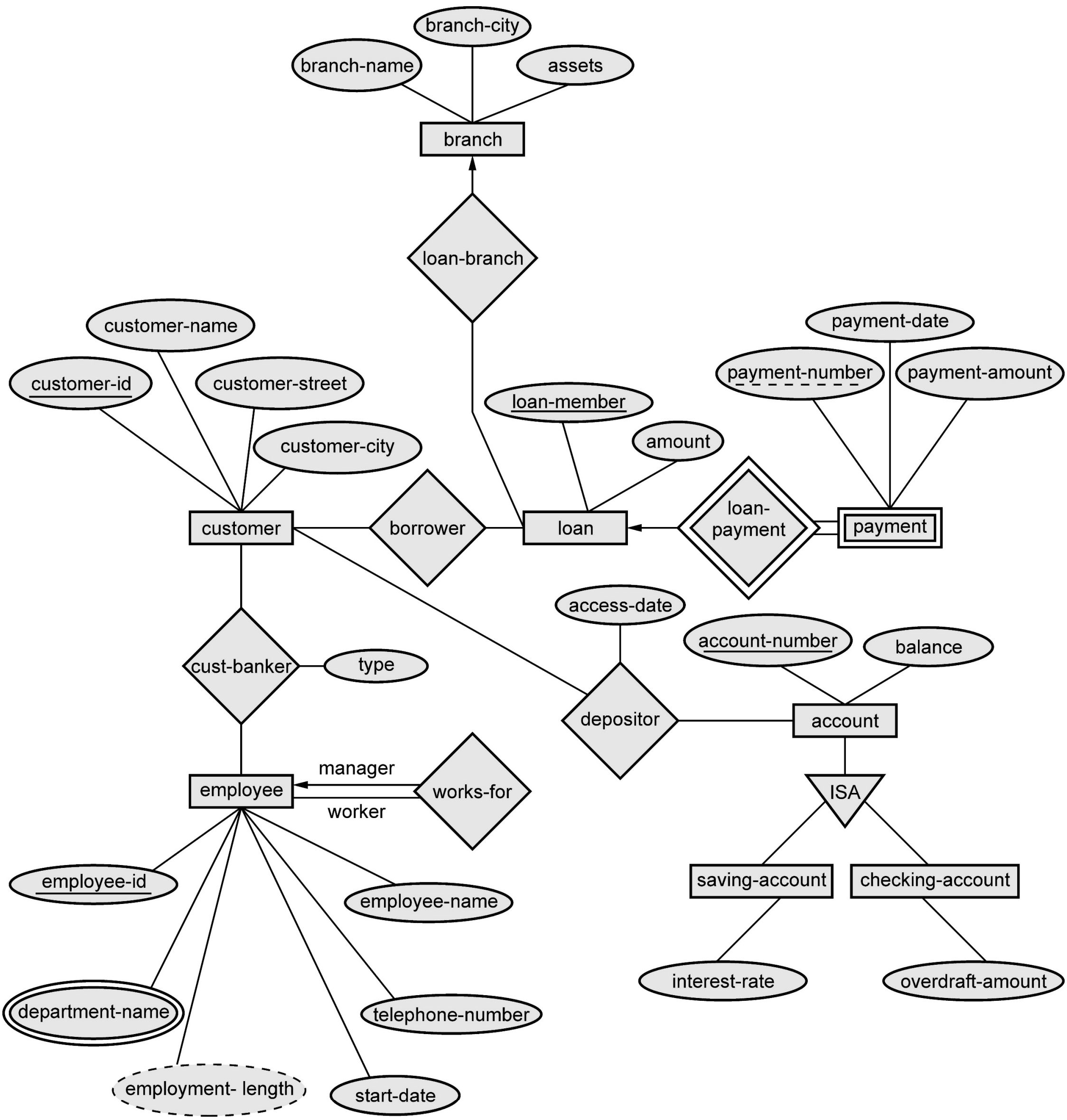
1. *Writing short note on – Mapping ISA relationship of E-R diagram to tables*

SPPU : Nov.-18, End Sem, Marks 4

1.21 Examples based on ER Diagram

Example 1.21.1 *Draw an ER diagram for banking system(Home-Loan Application)*

Solution :



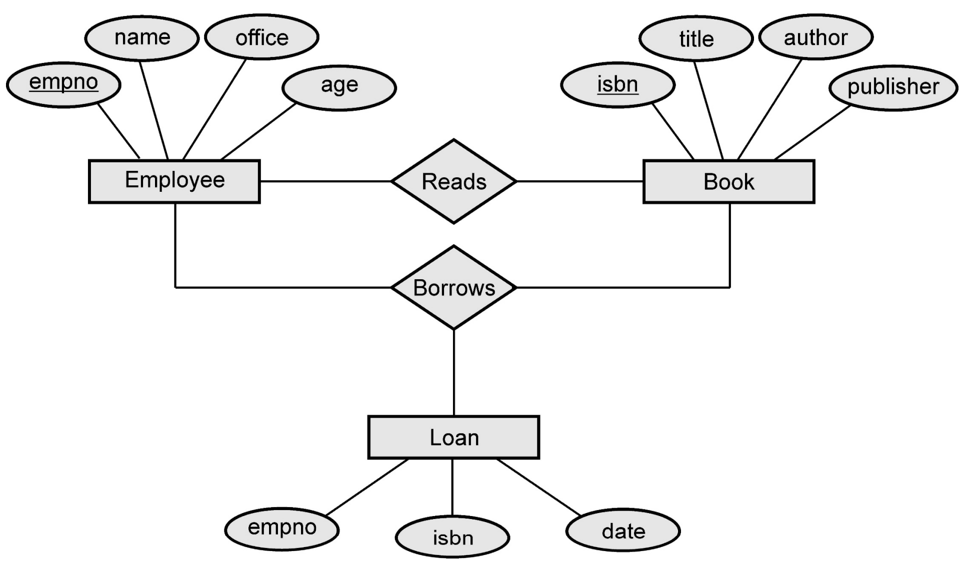
Example 1.21.2 *Consider the relation schema given in following Figure. Design and draw an ER diagram that capture the information of this schema.*

Employee(empno,name,office,age)

Books(isbn,title,authors,publisher)

Loan(empno,isbn,date)

Solution :



Example 1.21.3 *A car rental company maintains a database for all vehicles in its current fleet. For all vehicles, it includes the vehicle identification number license number, manufacturer, model, date of purchase and color. Special data are included for certain types of vehicles.*

Trucks : Cargo capacity

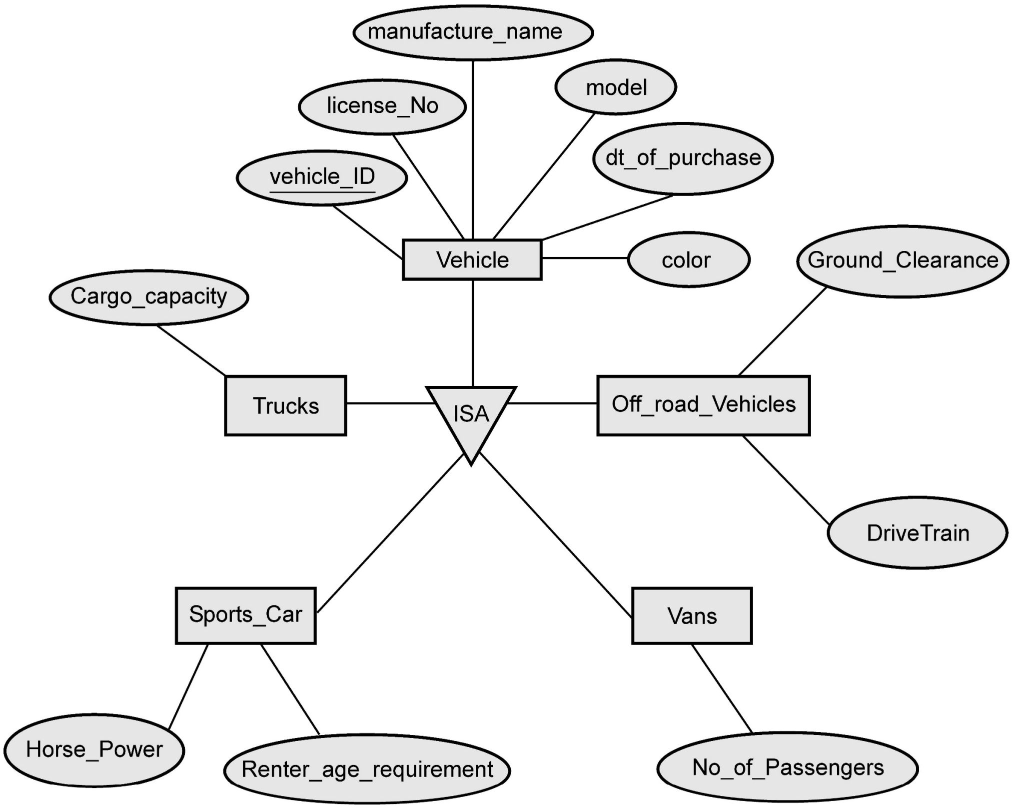
Sports cars : horsepower, renter age requirement

Vans : number of passengers

Off-road vehicles : ground clearance, drivetrain (four-or two-wheel drive)

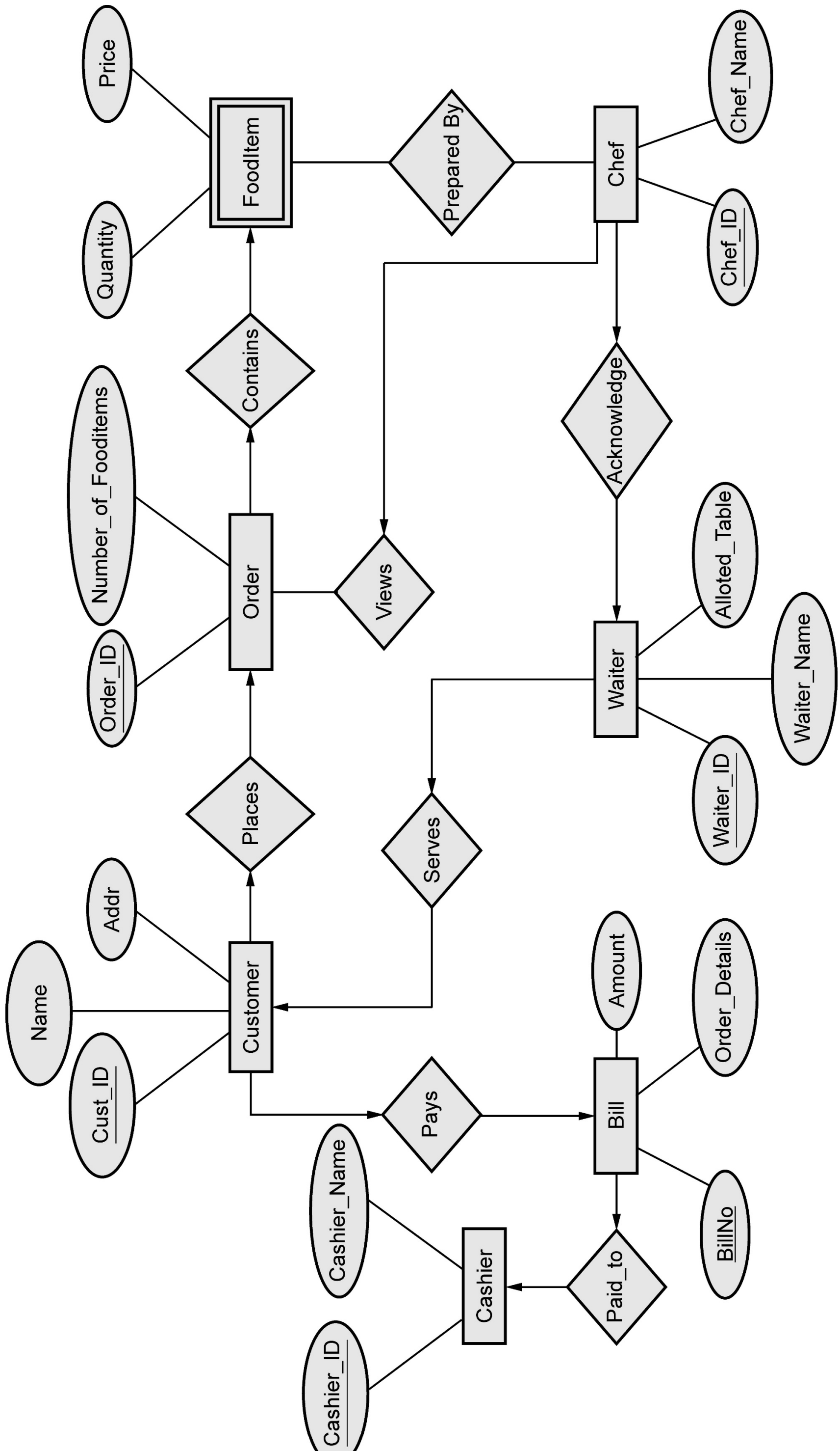
Construct an ER model for the car rental company database.

Solution :



Example 1.21.4 *Draw E-R diagram for the "Restaurant Menu Ordering System", which will facilitate the food items ordering and services within a restaurant. The entire restaurant scenario is detailed as follows. The customer is able to view the food items menu, call the waiter, place orders and obtain the final bill through the computer kept in their table. The Waiters through their wireless tablet PC are able to initialize a table for customers, control the table functions to assist customers, orders, send orders to food preparation staff (chef) and finalize the customer's bill. The Food preparation staffs (chefs), with their touch-display interfaces to the system, are able to view orders sent to the kitchen by waiters. During preparation they are able to let the waiter know the status of each item, and can send notifications when items are completed. The system should have full accountability and logging facilities, and should support supervisor actions to account for exceptional circumstances, such as a meal being refunded or walked out on.*

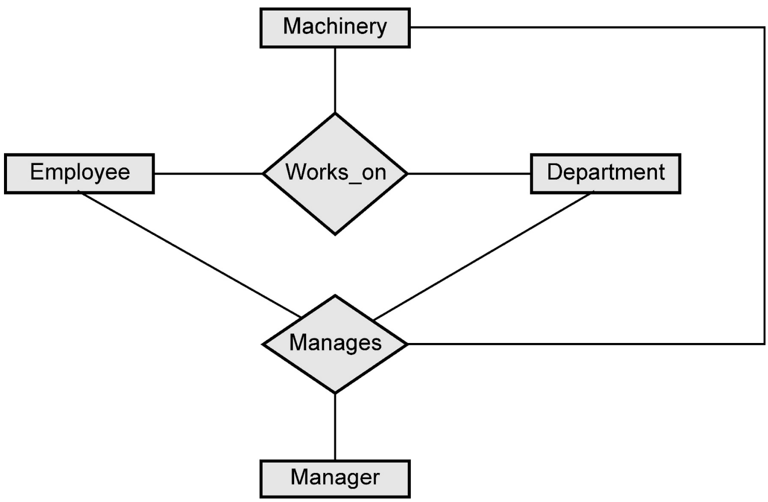
Solution :


Price
FoodItem
Quantity
Prepared By
Chef
Chef_Name
Chef_ID
Contains
Acknowledge
Number_of_Fooditems
Order
Views
Order_ID
Alloted_Table
Waiter
Waiter_Name
Waiter_ID
Places
Serves
Addr
Name
Customer
Cust_ID
Amount
Order_Details
Pays
Bill
BillNo
Cashier_Name
Cashier
Paid_to
Cashier_ID

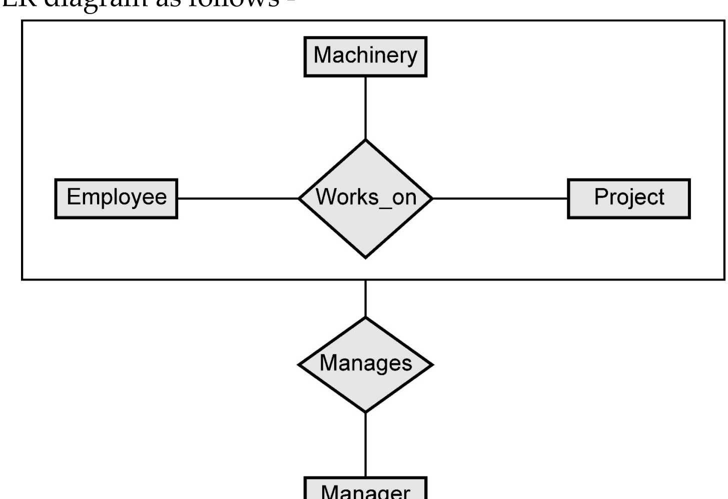
Example 1.21.5 *What is aggregation in ER model ? Develop an ER diagram using aggregation that captures following information : Employees work for projects. An employee working for particular project uses various machinery. Assume necessary attributes. State any assumptions you make. Also discuss about the ER diagram you have designed.*

Solution : Aggregation : Refer section 1.19.3.

ER Diagram : The ER diagram for above described scenario can be drawn as follows -



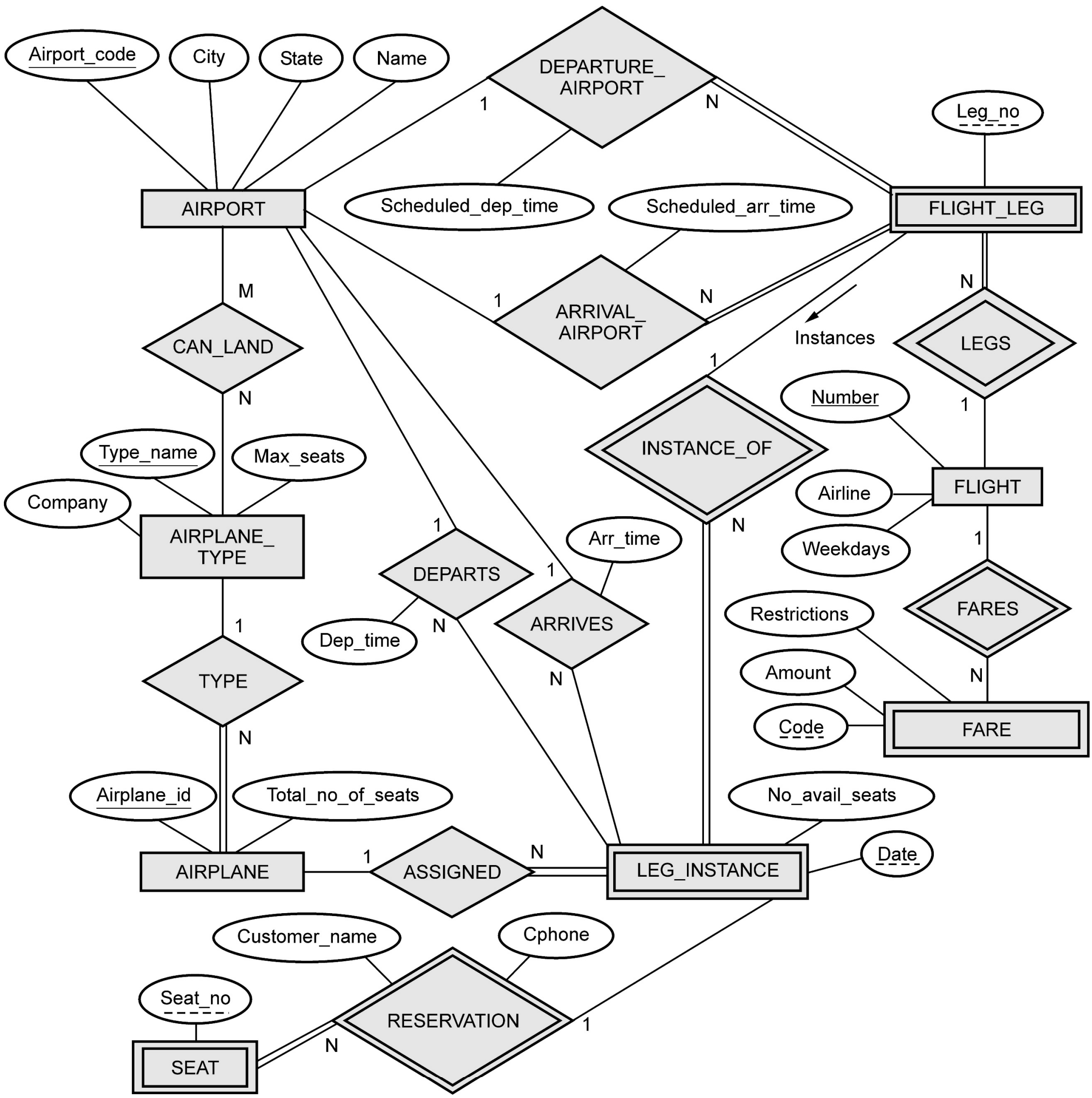
The above ER model contains the redundant information, because every Employee, Project, Machinery combination in **works_on** relationship is also considered in **manages** relationship. To avoid this redundancy problem we can make use of aggregation relationship in ER diagram as follows -



We can then create a binary relationship **manages** for between **Manager** and **(Employee, Project, Machinery).**

Example 1.21.6 *Draw an ER diagram of Airline reservation system taking into account at least five entities. Indicate all keys, constraints and assumptions that are made*

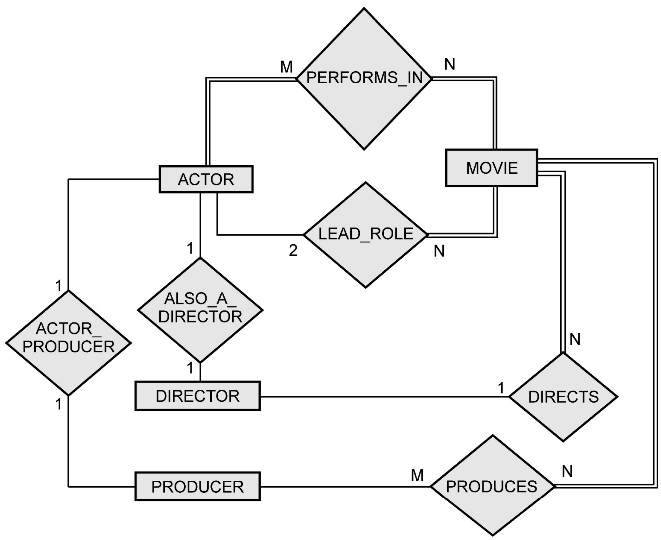
Solution :



A LEG is nonstop portion of flight. The LEG_INSTANCE is a particular occurrence of a LEG on particular date.

Example 1.21.7 *Draw an ER diagram of movie database. Assume your own entities (minimum 4) attributes and relationships.*

Solution :



Example 1.21.8 *Draw an ER diagram to represent the Election Information system based on the following description.*

In Indian national election, a state is divided into a number of constituencies depending upon the population of the state. Several candidates contest elections in each constituency. Record the number of votes obtained by each candidate. The system also maintains the voter list and a voter normally belongs to a particular constituency.

Note that the party details must also be taken care in the design.

Solution :

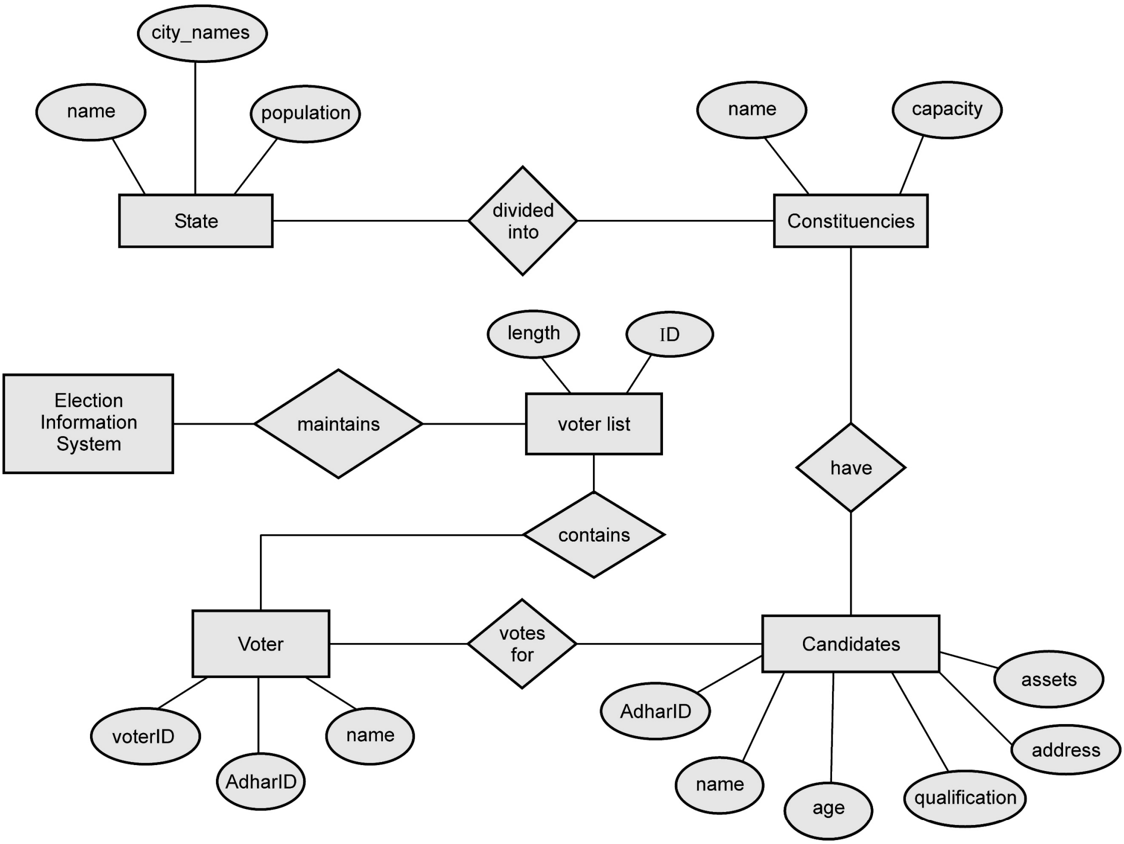
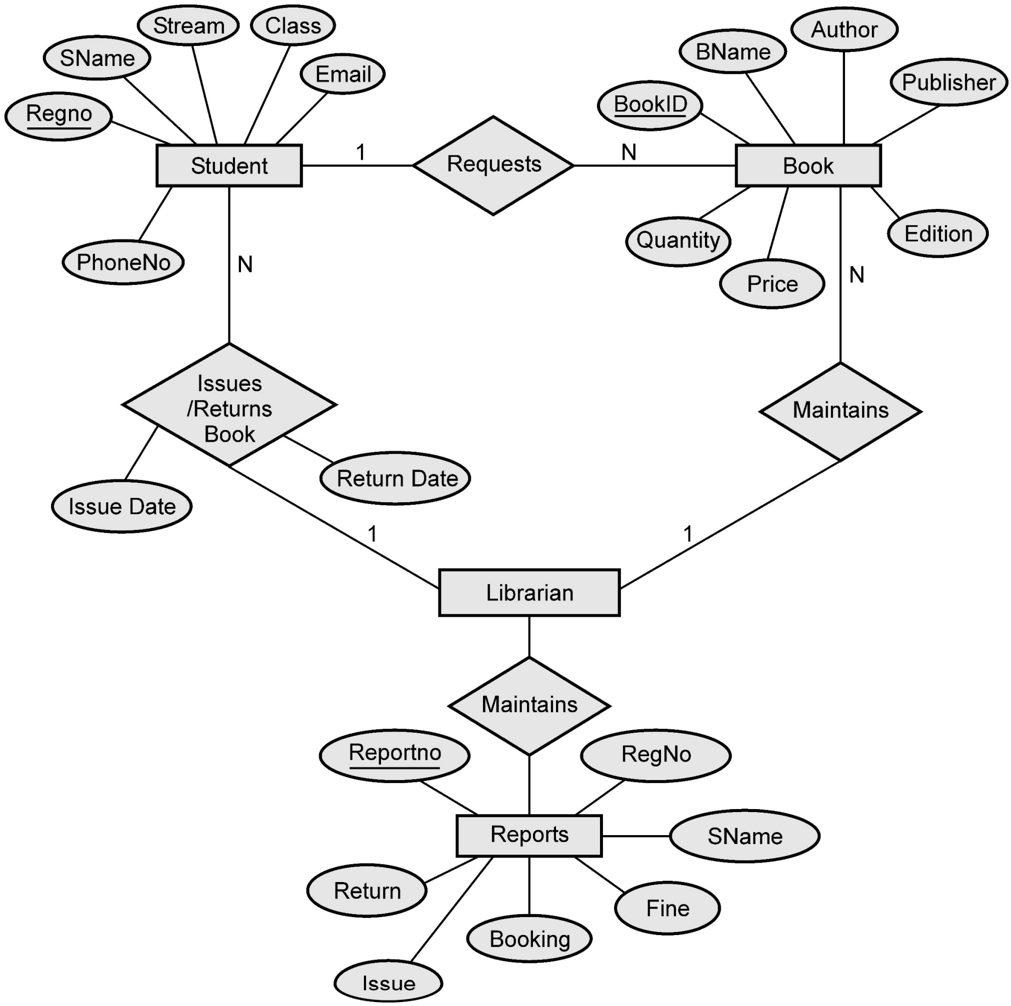


Fig. 1.21.1 Election information system

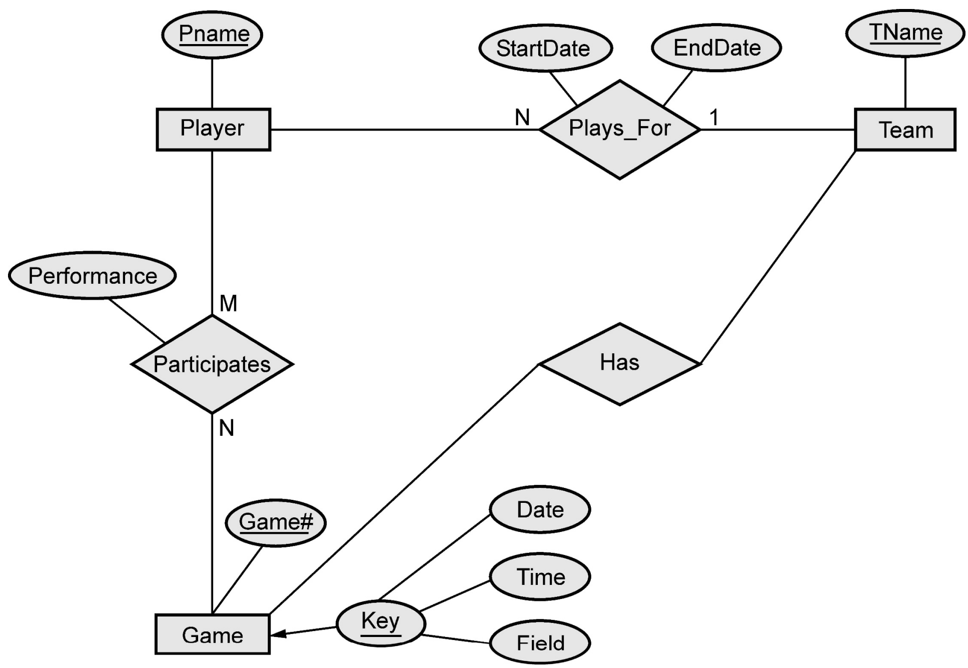
Example 1.21.9 *Construct an E R diagram for library management system*

Solution :



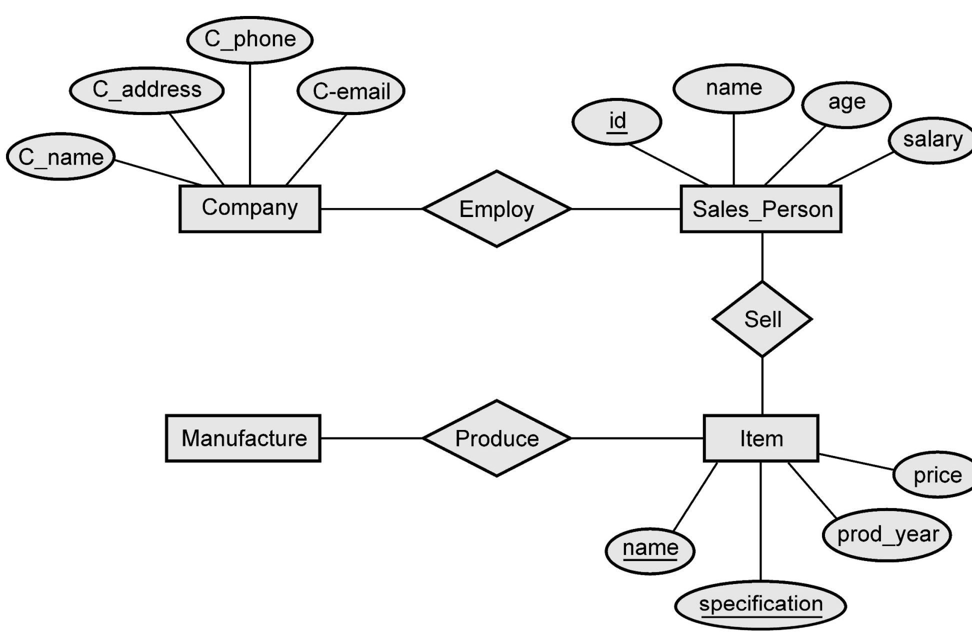
Example 1.21.10 *A database is to be constructed to keep track of the teams and games of a sport league. A team has number of players not all of whom participate in each game. It is desired to keep track of the players participating in each game for each team, the positions they played in that game, and the result of the game. Design an ER diagram completely with attributes, keys and constraints for the above description.*

Solution :



Example 1.21.11 *Draw an ER diagram for a small marketing company database. Assume suitable data.*

Solution :



Example 1.21.12 *A university database contains information about professors (identified by social security number, or SSN) and courses (identified by courseid). Professors teach courses; each of the following situations concerns the Teaches relationship set. For each situation, draw an ER diagram -*

1. Professors can teach the same course in several semesters, and each ordering must be recorded.

2. Professors can teach the same course in several semesters, and only the most recent such ordering needs to be recorded (Assume this condition applies in all subsequent questions).

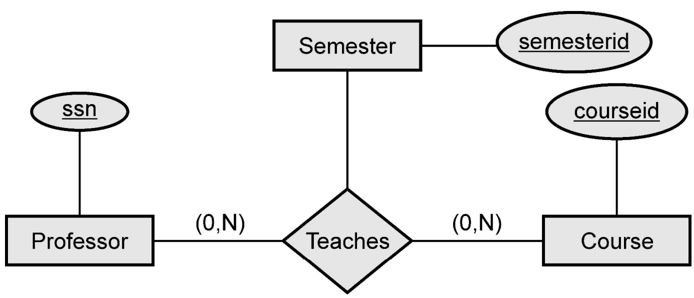
3. Every professor must teach some course.

4. Every professor teaches exactly one course.

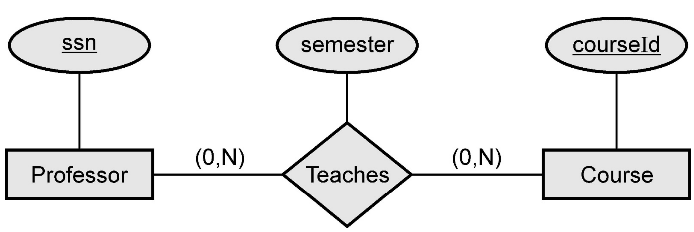
5. Every professor teaches exactly one course and every course must be taught by some professor.

Solution :

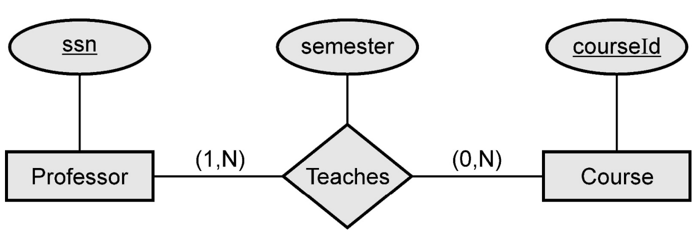
1.



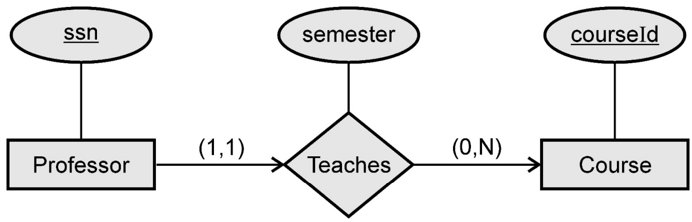
2.



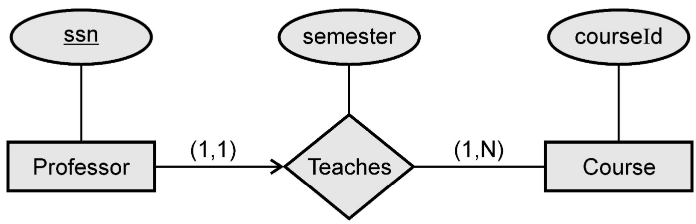
3.



4.

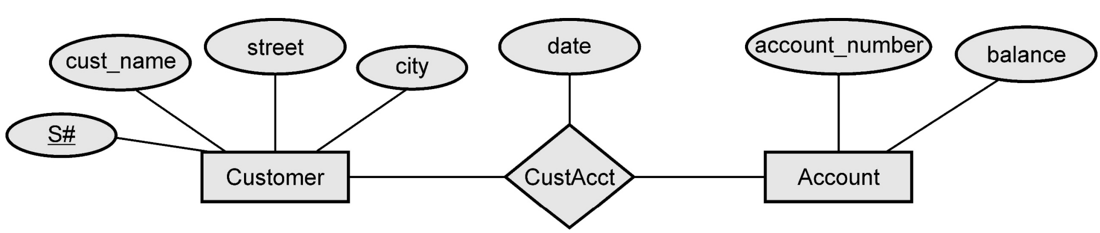


5.



Example 1.21.13 *Construct an ER diagram of customer account relationship. Customer entity with attributes of S#, Customer name, street, customer city, and account entity with attributes account no, and balance. The customer account relationship with date attributes.*

Solution :



Example 1.21.14 *A university registrar's office maintains data about the following entities :*

(1) courses, including number, title, credits, syllabus, and prerequisites;

(2) course offerings, including course number, year, semester, section number, instructor(s), timings, and classroom;

(3) students, including student-id, name, and program; and

(4) instructors, including identification number, name, department, and title.

Further, the enrollment of students in courses and grades awarded to students in each course they are enrolled for must be appropriately modeled.

(i) Construct an E-R diagram for the registrar's office. Document all assumptions that you make about the mapping constraints.

SPPU : Aug.-17, May-19, End Sem, Marks 10

OR

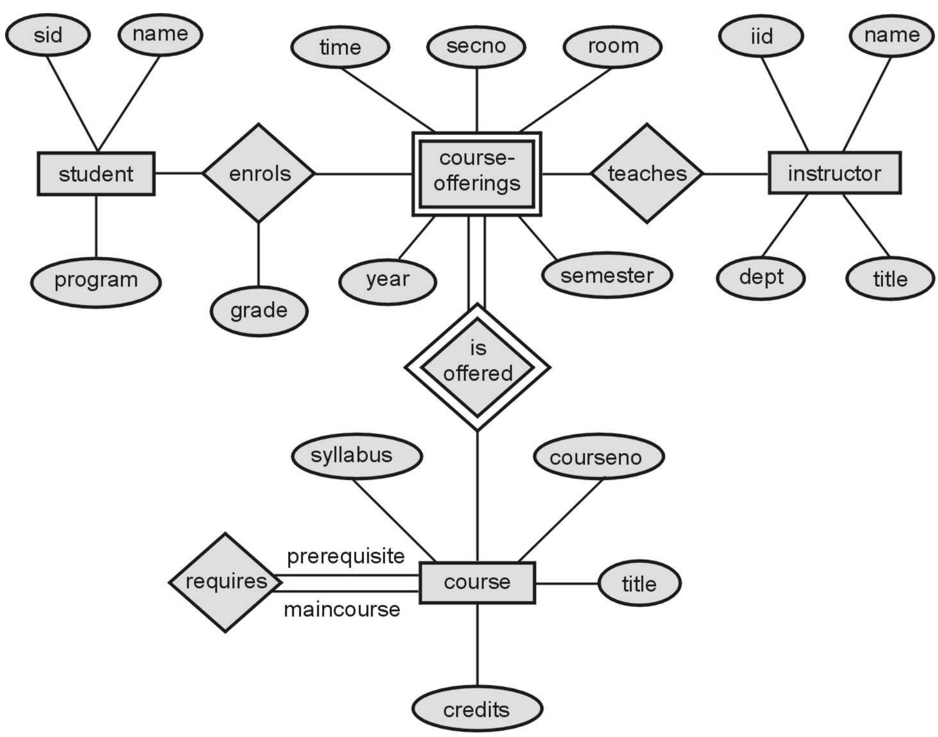
Consider a database used to record the marks that students get in different exams of different course offerings.

Construct an E - R diagram that models exams as entities, uses a ternary relationship, for the above database.

SPPU : Dec.-17, End Sem, Marks 5

Solution :

(i)



Example 1.21.15 *Assuming you as a part of designer and development team, propose E-R model using E-R diagram for the following data requirements of banking system. Also convert and represent E-R model into tables characteristics of banking requirements are given below.*

The bank is organized into branches. Each branch is located in a particular city.

Bank customers are identified by customer_id.

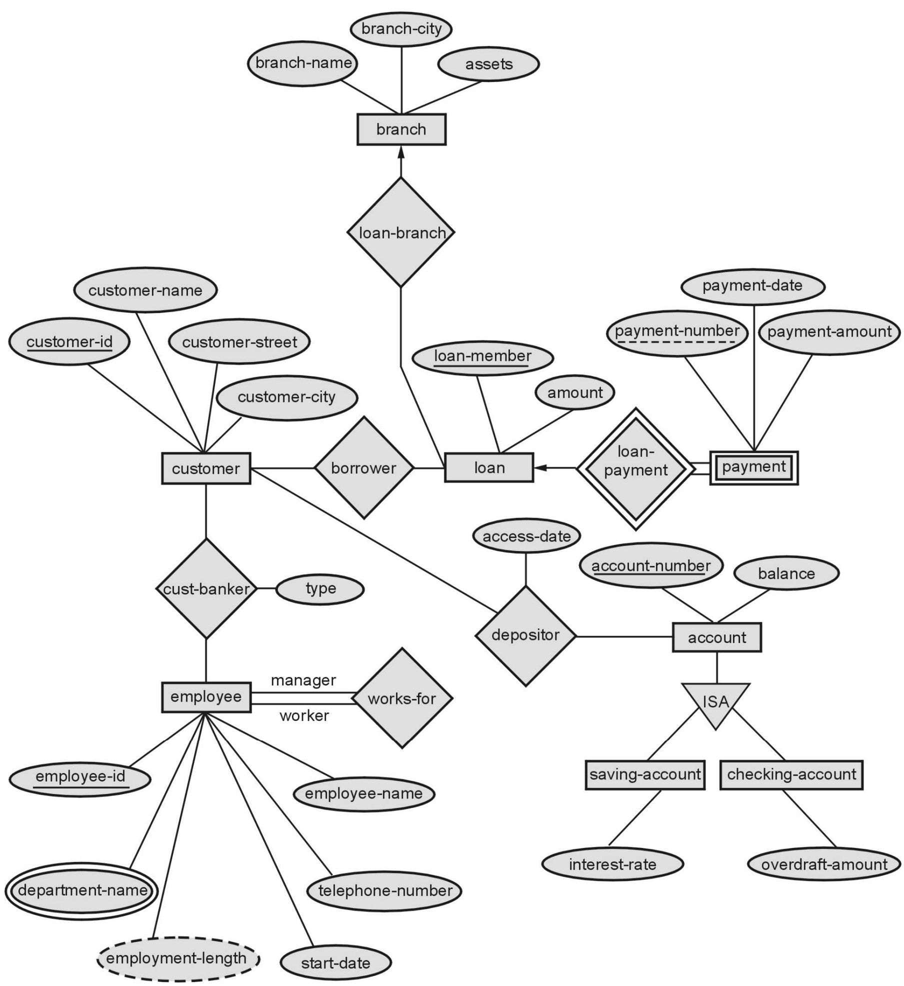
Bank employees are identified by emp-id

The bank offers two types of accounts saving and current. Accounts can be held by more than one customer and a customer can have more than one account.

A loan originates at a particular branch and can be held by one or more customers.

Additional requirement can be assumed if required, but assumptions should be clearly mentioned. **SPPU : Oct.-18, In Sem, Marks 10**

Solution :



Example 1.21.16 *Ramesh's family owns and operates a 100-acre farm for several generations. Since the farm business is growing, Ramesh is thinking to build a database that would make easier the management of the activities in the farm. He is considering the following requirements for the database :*

i) For each livestock classification group (for example, cow, horse etc.), Ramesh keeps track of the following: identification number, classification, total number of livestock per classification group (for example, number of cows, number of horses etc.)

ii) For each crop the following information is recorded Crop identification number and classification.

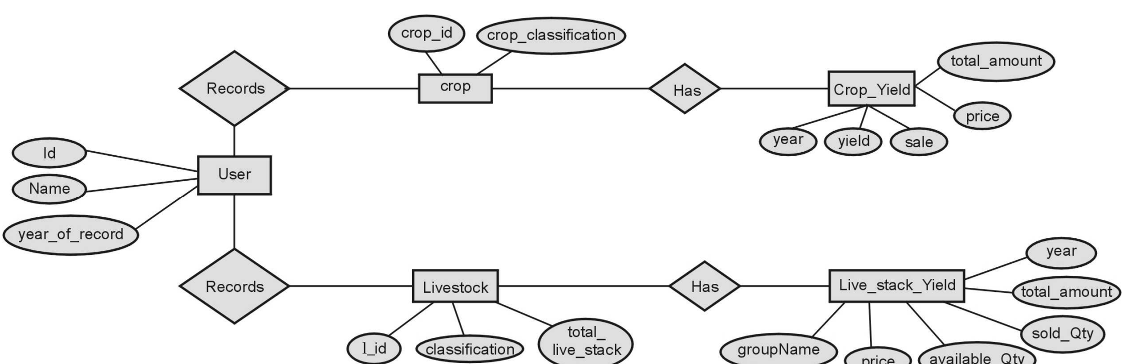
iii) Ramesh has recorded the yield of each crop classification group during the last ten years. The records consist of the year, yield, sales, price of the crop and the amount of money earned.

iv) Ramesh has recorded the yield of each livestock classification group during the last ten years. The records consist of the following historical data : the year, (historical) selling price per head, number of livestock in the end of the year, number of livestock sold during one-year period, and the total amount of money earned.

Draw an E-R diagram for this application. Specify the key attribute of each entity type.

SPPU : Dec.-18, End Sem, Marks 5

Solution :



Example 1.21.17 *Assuming you as a part of design and development of team of an organization, propose E-R model using E-R diagram for the following data requirements. Also convert and represent E-R model into tables :*

Company organized into DEPARTMENT. Each department has unique name and a particular employee who manages the department. Start date for the manager is recorded. Department may have several locations.

A department control a number of PROJECT. Projects have a unique name, number and a *single location.*

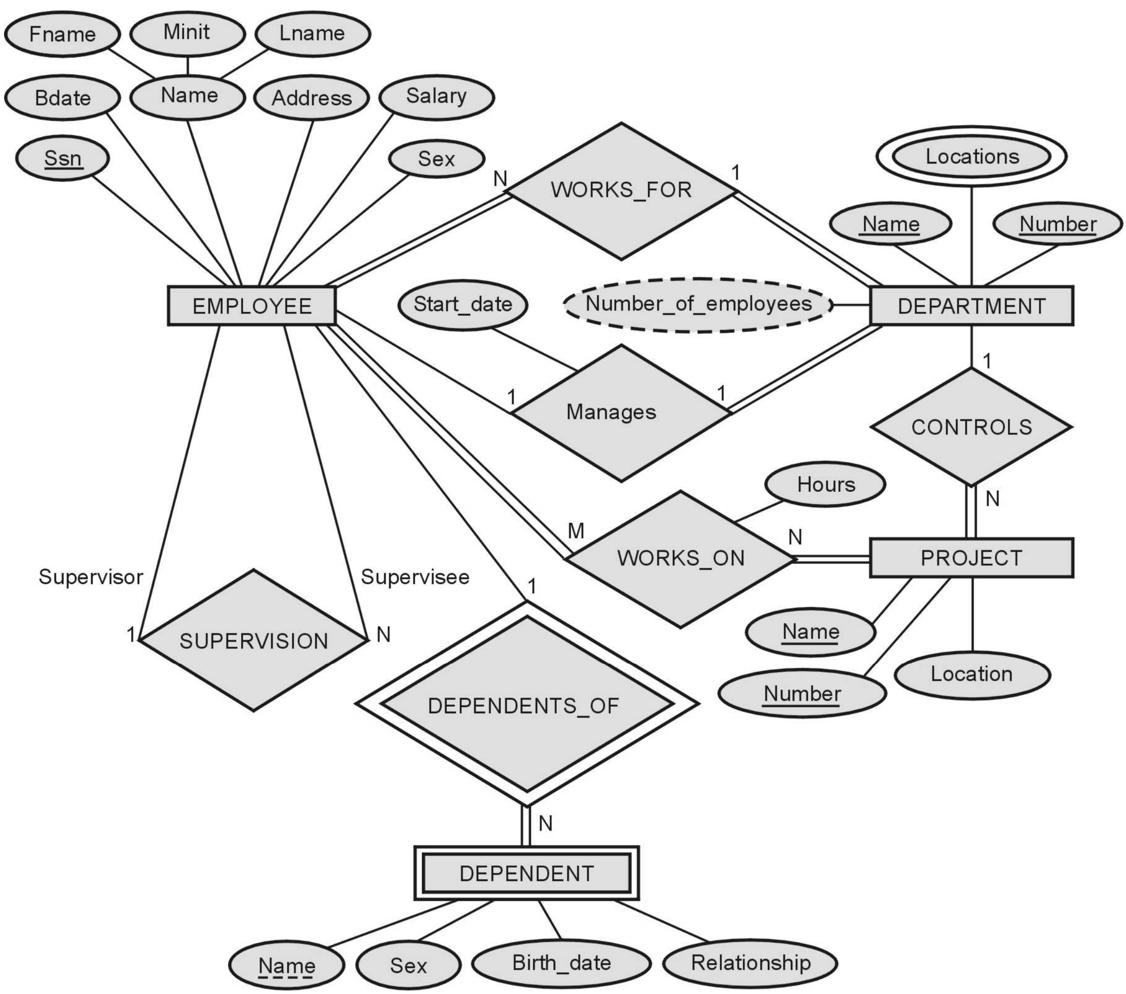
Company's EMPLOYEE name, ssno, address, salary, sex and birth date are recorded. An employee is assigned to one department, but may work for several projects (not necessarily controlled by her dept).

Number of hours/week an employee works on each project is recorded.

Employee's DEPENDENT are tracked for health insurance purposes (dependent name, birthdate, relationship to employee).

SPPU : May-19, End Sem, Oct.-19, In Sem, Marks 10

Solution :



The above ER diagram is converted into tables as

EMPLOYEE

ssn	Fname	Lname	Minit	Bdate	Address	sex	salary

DEPARTMENT

DNum	DName	Mng_ssn	Mng_start_date

DEPT_LOCATIONS

Dept_num	Dept_location

PROJECT

Proj_Name	Proj_Num	Proj_Location	DNo

WORKS_ON

Emp_SSN	Proj_No	Hours

DEPENDENT

Emp_SSN	Depend_Name	Sex	Bdate	Relationship

Example 1.21.18 *Assume we have the following application that models soccer teams, the games they play and the players in each team. In the design, we want to capture the following :*

We have a set of teams, each team has an ID (unique identifier), name, main stadium and to which city this team belongs.

Each team has many players and each player belongs to one team. Each player has a number (unique identifier), name, DoB, start year and shirt number that he uses.

Teams play matches, in each match there is a host team and a guest team. The match takes places in the stadium of the host team.

For each match we need to keep track of the following :

The date on which the game is played.

The final result of the match.

The players participated in the match. For each player, how many goals he scored, whether or not he took yellow card and whether or not he took red card.

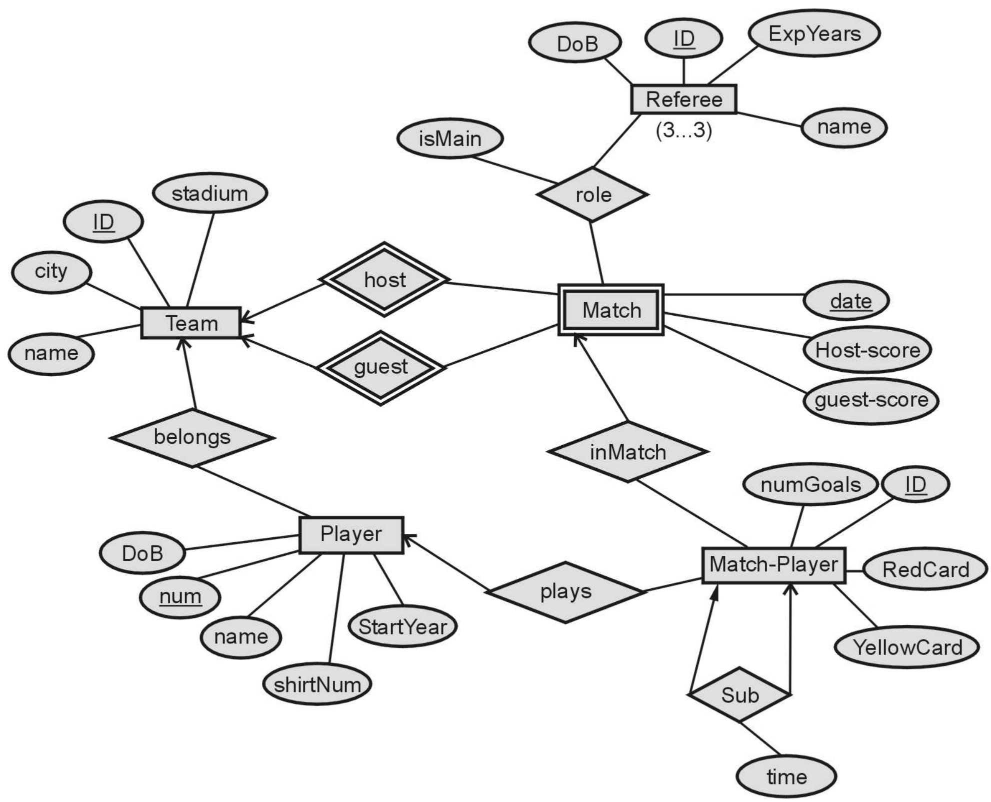
During the match, one player may substitutee another player. We want to capture this substitution and the time at which it took place.

Each match has exactly three referees. For each referee we have an ID (unique identifier), name, DoB, years of experience. One referee is the main referee and the other two are assignment referee.

Design an ER diagram to capture the above requirements. State any assumptions you have that effects your design.

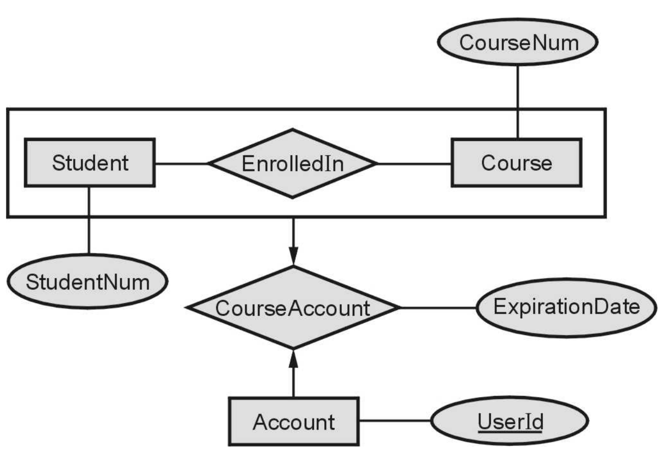
SPPU : Dec.-19, End Sem, Marks 5

Solution :



Example 1.21.19 *Translate the following entity-relationship diagram to relational tables.*

SPPU : Dec.-19, End Sem, Marks 5



Solution :

Student

StudentNum

Course

CourseNum

Account

UserId

EnrolledIn

StudentNum	CourseNum

CourseAccount

UserId	StudentNum	CourseNum	ExpirationDate

Example 1.21.20 *A weak entity set can always be made into strong entity set by adding to its attributes; the primary key attributes of its identifying entity set. Outline what sort of redundancy will result if we do so while converting into tables.*

SPPU : Aug.-17, Oct.-18, In Sem, Marks 5

Solution : By adding appropriate primary key to a weak entity set can result into the strong entity set.

If we add primary key attributes to the weak entity set, they will be present in both the entity set and the relationship set and both needs to be the same. Hence there will be redundancy.

Unit - I
Multiple Choice Questions

Q.1 A DBMS provides users with the conceptual representation of _______.

a register b data

c logical view d physical view

Q.2 DBMS helps to achieve _______.

a data Independence b centralized Control of Data

c neither a nor b d both a and b

Q.3 In view of total database content is _______.

a conceptual view b internal view

c external view d physical view

Q.4 The main purpose of DBMS is to provide _____ view of data to user.

a completer b abstract

c partial d none of these

Q.5 ___ means to hide certain details of how data is stored.

a Data Integrity b Data independence

c Data abstraction d Data separation

Q.6 How many levels of data abstraction are there ?

a One b Two

c Three d Four

Q.7 A _____view of data expresses the way a user thinks about data _______.

a logical view b physical view

c both d none

Q.8 A physical view of data refers to the way data is handled at a________ its storage and retrieval.

a high level b low level

c medium level d all of these

Q.9 Architecture of the database can be viewed as _______.

a two levels b three levels

c four levels d one level

Q.10 In the architecture of a database system external level is the _______.

a physical level
b logical level
c conceptual level
d view level

Q.11 In hierarchical model records are organized as _______.

a lists
b links
c tree
d graph

Q.12 There are ____ levels of data independence.

a one
b two
c three
d four

Q.13 The ability to modify the schema of database in one level without affecting the schema definition in higher level is called as _______.

a data isolation
b data abstraction
c data hiding
d data independence

Q.14 Which of the following is record based on logical model ?

a Network Model
b Object Oriented Model
c E-R Model
d None of these

Q.15 The DDL is used to specify the_____.

a conceptual schemas
b internal schemas
c both
d none

Q.16 DCL stands for ______.

a Data Control Language
b Data Console Language
c Data Console Level
d Data Control Level

Q.17 Which of the following is / are the DDL statements ?

a Create
b Drop
c Alter
d All of the above

Q.18 Which are the three levels of abstraction ?

a Physical
b Logical
c External
d All of these

Q.19 The statement in SQL which allows to change the definition of a table is ___________.

a create
b alter
c select
d update

Q.20 Which of the following is NOT a basic element of all versions of the E - R model ?

a Entities
b Relationships
c Attributes
d Primary key

Q.21 Data independence means ______.

a data is defined separately and not included in programs
b programs are not dependent on the physical attributes of data.
c programs are not dependent on the logical attributes of data
d both (b) and (c)

Q.22 E-R model uses this symbol to represent weak entity set ______.

a dotted rectangle
b diamond
c doubly outlined rectangle
d none of these

Q.23 ______ express the number of entities to which another entity can be associated via a relationship set.

a Mapping cardinality
b Relational cardinality
c Participation constraints
d None of the mentioned

Q.24 In E-R diagram derived attribute is represented by ______.

a rectangle
b circle
c dashed Ellipse
d diamond

Q.25 DBA stands for ______.

a Data Building Administrator
b Database Access
c Database Authentication
d Database Administrator

Q.26 ______ represents the number of entities to which another entity can be associated

a Degree
b Cardinality
c Modality
d None of these

Q.27 Data Model is collection of conceptual tools for describing ______.

a Data
b Schema
c constraints
d All of the above

Q.28 Which of the following is example of Object based logical model ?

a Relational Model
b Hierarchical Model
c Network Model
d Entity Relationship Model

Q.29 Entity Relationship model consists of collection of basic objects called ______ and relationship among these objects.

a functions | b models
c entity | d all of these

Q.30 ER model was introduced by _____.

a E.F. Codd | b P.P. Chen
c Bjarne Stroustrup | d None of these

Q.31 ER diagram an ellipse represents _____.

a weak entity | b relationship
c attribute | d entity class

Q.32 In ER diagram the relationship is represented by _____.

a rectangle | b ellipse
c diamond | d circle

Q.33 Relationship among entities of a single class is called as _____.

a IS-A relationship | b recursive relationship
c HAS -A relationship | d none of these

Q.34 The relationship used for connecting entities of different types when identifiers are different _____.

a HAS-A relationship | b IS-A relationship
c recursive relationship | d none of these

Q.35 Which is not an example of strong entity type ?

a Employee | b Department
c Emp_ID | d Course

Q.36 If person is an entity then "Ankita" and "Prajkta" is the entity _____.

a characteristics | b field
c identifier | d instance

Q.37 Properties that describe the characteristics of entities are called :

a entities | b attributes
c identifiers | d relationships

Q.38 A student RollNumber, Name and Marks are all examples of _____.

a entities | b attributes
c relationships | d none of these

Q.39 An attribute which consists of a group of attributes is called as _____.

a composite attribute | b single valued attribute
c multi-valued attribute | d none of these

Q.40 The attribute "age" is calculated from "date_of_Birth". The attribute "age" is _____.

a single valued
b multi valued
c composite
d derived

Q.41 The most common type of relationship used in data modeling is _____.

a unary
b binary
c ternary
d none of these

Q.42 An entity type whose existence depends on another entity type is called ___ entity.

a strong
b weak
c dependent
d mutually dependent

Q.43 Every weak entity set can be converted into a strong entity set by :

a Using generalization
b Adding appropriate attributes
c Using aggregation
d None of the above

Q.44 In a one-to-many relationship, the entity that is on the one side of the relationship is called a(n) ________ entity.

a parent
b child
c instance
d subtype

Q.45 Which of the following is NOT a basic element of all versions of the E-R model ?

a Entities
b Attributes
c Relationships
d Primary keys

Q.46 The entity employee has three candidate keys : 1) EmpID 2) Email 3) Date_of_joining 4) Designation. Suggest best primary key for this entity :

a EmpID
b Email
c Date_of_joining
d Designation

Answers Keys for Multiple Choice Questions :

Q.1	b	**Q.2**	d	**Q.3**	a	**Q.4**	b
Q.5	c	**Q.6**	c	**Q.7**	a	**Q.8**	b
Q.9	b	**Q.10**	d	**Q.11**	c	**Q.12**	b
Q.13	d	**Q.14**	a	**Q.15**	a	**Q.16**	a
Q.17	d	**Q.18**	d	**Q.19**	b	**Q.20**	d

Q.21	d	**Q.22**	c	**Q.23**	a	**Q.24**	c
Q.25	d	**Q.26**	b	**Q.27**	d	**Q.28**	d
Q.29	c	**Q.30**	b	**Q.31**	c	**Q.32**	c
Q.33	b	**Q.34**	a	**Q.35**	c	**Q.36**	d
Q.37	b	**Q.38**	b	**Q.39**	c	**Q.40**	d
Q.41	b	**Q.42**	b	**Q.43**	b	**Q.44**	a
Q.45	d	**Q.46**	a				

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Notes